

Transforming Challenges into Sustainable Solutions

Annual & Sustainability Report 2023/2024



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● Vision

A clean environment, towards a liveable and sustainable Singapore

● Mission

To ensure a clean and sustainable environment for Singapore, together with our partners and the community

● Roles

- Maintain high standards of public health
- Improve and sustain a clean environment
- Promote sustainability and resource efficiency
- Provide timely and reliable meteorological services
- Encourage a vibrant hawker culture

● Core Values



Empathy

Care for the safety and well-being of our people and stakeholders, and serve them with empathy



Agility

Embrace change and innovation



Resilience

Show courage and fortitude in the face of challenges



Teamwork

Collaborate and co-create with our colleagues and stakeholders



Heart for the Environment

Safeguard the environment with passion and inspire others to do the same

● ABOUT THIS REPORT

In this Annual & Sustainability Report, the National Environment Agency (NEA) provides an account of our corporate, financial, and sustainability performance in Financial Year (FY) 2023 – from 1 April 2023 to 31 March 2024. We hope this report will serve as a useful reference for other organisations and agencies, and inspire them to embark on their own journey of sustainability.

REPORTING SCOPE¹

This report, published in September 2024², documents NEA's organisational activities, achievements, and financial performance in FY2023 across our key premises, unless otherwise stated.

Stand-alone Office Premises

- Centre for Climate Research Singapore at Kim Chuan Road
- Choa Chu Kang Cemetery Office at Old Choa Chu Kang Road
- NEA Regional Offices:
 - Central Regional Office at Jalan Bukit Merah
 - Eastern Regional Office at Tannery Lane
 - Eastern Regional Office (Satellite Office) at Sin Ming Drive
 - Western Regional Office at Albert Winsemius Lane

Tenanted Office Premises

- Changi Meteorological Station
- Environmental Health Institute at Ang Mo Kio Tech Place II
- Environmental Health Institute at Biopolis
- Environmental Health Institute at Van Kleef Centre
- Hawker Centres Group at HDB Hub
- Meteorological Service Singapore at Changi Airport
- National Radiochemistry Laboratory at CREATE Tower
- NEA Headquarters at Environment Building
- Pollution Control Division 1 and 2 at Bukit Merah Central
- Secondary Standards Dosimetry Laboratory at Health Sciences Authority

- Seletar Meteorological Station
- Singapore Environment Institute at Lifelong Learning Institute

Industrial Premises

- Semakau Landfill
- Tuas Marine Transfer Station
- Tuas South Incineration Plant

Other Premises

- Burial Preparation Facility (Jalan Bahar)
- Choa Chu Kang Cemetery, Columbarium, and Crematorium
- Garden of Peace @ CCK (Inland Ash Scattering Facility)
- Hawker Centres and Markets³
- Kranji State Cemetery
- Mandai Crematorium and Columbarium
- Yishun Columbarium

ASSURANCE

External assurance was not sought for the reporting period. Nonetheless, the data included in this report had undergone rigorous internal processes, such as data monitoring and review, to ensure accuracy.

ENQUIRIES

As part of NEA's environmental conservation efforts, the report will not be printed. Digital copies of past and current reports can be found on NEA's corporate website: www.nea.gov.sg/publications

For feedback or enquiries, please contact the NEA Corporate Communications Division: Ms Eleanor Goh: eleanor_goh@nea.gov.sg

¹ This report's annual and financial reporting scope covers all NEA premises. Our sustainability reporting scope aligns with GreenGov.SG, which covers premises whose resource consumption NEA has operational control over. In this regard, NEA's resource consumption performance excludes the National Radiochemistry Laboratory at the Campus for Research Excellence and Technological Enterprise (CREATE), as the resource consumption data is aggregated for entire CREATE Tower, and breakdown of data for individual laboratories is unavailable

² The forward-looking statements in this report are accurate as at September 2024, and will not be updated to reflect events or circumstances that may arise after the date of publication

³ For the full list of hawker centres and markets, visit www.nea.gov.sg/our-services/hawker-management



● CHAIRMAN'S FOREWORD



Mr Lee Chuan Seng
Chairman

2023 stands as the warmest year on record globally and the fourth warmest in Singapore, underscoring the pressing reality of the climate crisis.

As a small, low-lying city state, Singapore is particularly susceptible to the effects of climate. Earlier this year, NEA released the Third National Climate Change or V3 Study, which projected higher temperatures, more extreme weather patterns, and higher sea level rises in the decades to come. To better prepare for hotter weather, NEA and the Ministry of Sustainability and the Environment (MSE) introduced the heat stress advisory to advise the public on precautions to be taken while outdoor on hot days. At the same time, we are redoubling efforts to achieve net zero, such as through the launch of the enhanced Climate Friendly Households Programme to encourage the purchase of energy- and water-efficient products.

Sustainability goes beyond reducing our carbon footprint, to also include reducing consumption and improving recycling to build a circular economy. Waste generated per capita decreased over the past decade, but the recycling rate also came down. We are doing more to encourage reducing and recycling, such as the Bloobox initiative and the implementation of the disposable carrier bag charge last year. To complement these efforts, we are also exploring innovative solutions to repurpose the ash stored at Semakau Landfill as land reclamation infill.

As we build a more sustainable future, we must continue to keep dengue low. In 2023, Singapore averted a dengue outbreak through stepping up mosquito breeding inspections and expanding Project *Wolbachia*. As at March 2024, Project *Wolbachia* covered nearly 500,000 households. But Project *Wolbachia* is not a silver bullet, and all of us have a part to play to prevent mosquito breeding and keep dengue at bay.

Another area where everyone has a part to play is public hygiene. Overall satisfaction with public cleanliness remains high. We are grateful that the large majority have contributed to this effort, especially in returning their trays at hawker centres and coffeeshops. The tray return rate is now around 93 per cent, compared to the 65 per cent three years ago when table littering enforcement started.

With 2024 designated as the Year of Public Hygiene, NEA will intensify efforts to reinforce good hygiene practices. We will continue to partner and work with the community and other stakeholders to build a cleaner, greener, and more sustainable Singapore.



● CEO'S MESSAGE

Our mission to build a clean and sustainable environment for Singapore has become increasingly urgent. With 2023 being the hottest year on record globally, coupled with rapid technological advancements and an evolving social compact, we are accelerating our efforts to achieve our goals.

NEA is growing our capabilities in new areas. As the carbon regulator, we outlined the eligibility criteria for international carbon credits that can be used to partially offset taxable emissions from this year. In addition, we are growing our regulatory expertise in emerging areas such as carbon capture, utilisation, and storage. Another area is in developing a better understanding of the safety of nuclear energy, especially with the emergence of more advanced technologies and small modular reactors.

We are also transforming and making continuous improvements to our existing areas of work, especially in the areas of cleanliness, waste management operations, after-death, enforcement, and regulations. This will enable us to achieve better outcomes and improved service delivery, while making the most of technology and bringing out the best in our workforce. We do so by forming stronger partnerships with industry and the community. In designating this year as the Year of Public Hygiene, we are rallying everyone to make Singapore a cleaner and more liveable city. We are grateful for all your support in the many different areas that you have contributed to.

At the heart of NEA is our people, and we are dedicated to putting them at the centre of what we do in every way that we can. This includes creating more opportunities for learning and upgrading, as well as a greater emphasis on career development. Last year, we introduced a corporate bonus that would enable officers to better share in the fruits of our collective effort and to better recognise teamwork. As our work is very diverse, we are also looking to better recognise our officers for their deep skills and experience, as well as the complexity of the environment that they work in. The well-being of our officers and contractors remains a top priority, and we continued to strengthen our capabilities in workplace safety and health (WSH). We expanded our in-house WSH team to form the WSH Division, and we will be embarking on the bizSAFE3 certification this year.

Lastly, I would like to thank all NEA officers, past and present, for living our EARTH values of empathy, agility, resilience, teamwork, and heart for the environment. In particular, I would like to thank two members of the senior leadership team who have since retired – Director-General of the Meteorological Service Ms Wong Chin Ling, and Senior Advisor Mr Khoo Seow Poh. The dedication of all our officers, together with the strong support of industry partners and the community, will enable us to build a beautiful, cleaner, and greener Singapore for present and future generations.

Mr Wong Kang Jet
Chief Executive Officer



● BOARD OF DIRECTORS



Mr Lee Chuan Seng

Chairman,
National Environment
Agency



Mr Chaly Mah

Deputy Chairman, National
Environment Agency
Chairman, Surbana Jurong
Pte Ltd
Chairman, Netlink NBN Trust
with effect from 1 April 2024



Mr Wong Kang Jet

Chief Executive Officer,
National Environment
Agency



Mr Quek Suan Kiat

Former Vice Chairman,
Barclays Bank PLC Singapore



Mr Lim Eng Hwee

Chief Executive Officer,
Urban Redevelopment
Authority



Mr Puah Kok Keong

Chief Executive, Energy
Market Authority



Dr Andrew Khoo

Former Deputy Managing
Director, Monetary Authority
of Singapore



Ms Janet Young

Managing Director &
Group Head, Channels &
Digitalisation, Strategic
Communications & Brand,
United Overseas Bank Ltd



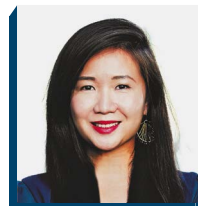
Mr Zia Zaman

General Manager, Corporate
Business Development Asia,
Microsoft



Mr Walter Fernandez

Editor-in-Chief and Chief
Sustainability Officer,
Mediacorp Pte Ltd



Ms Fang Eu-Lin

Partner, Sustainability and
Climate Change Practice
Leader, PwC Singapore



**Mr Suhaimi
Zainul-Abidin**

Chief Executive Officer,
Quantedge Capital Pte Ltd
till 31 March 2024



Ms Sylvia Choo

Director, Unions, National Trades
Union Congress
Executive Secretary, Singapore
Industrial and Services Employees'
Union
Co-General Manager, Tripartite
Alliance for Dispute Management
till 31 March 2024



Ms Susan Chong

President, Greenpac (S) Pte
Ltd
Chief Executive Officer,
Greenphyto
till 31 March 2024



Dr Azlinda Anwar

Executive Director,
Biomedical Research Council
Agency for Science,
Technology and Research
with effect from 1 April 2024



Ms Tan Man Ee

Chief Operating Officer &
Director, NatSteel Holdings
Pte Ltd
Chairperson, WSH Council
(Manufacturing) Committee
with effect from 1 April 2024



Ms Teo Siew Pan

Executive Secretary,
Advanced Manufacturing
Employees' Union
with effect from 1 April 2024



SENIOR LEADERSHIP TEAM

As at 1 September 2024



Mr Wong Kang Jet
Chief Executive Officer



Mr Ananda Ram Bhaskar
Deputy Chief Executive Officer,
Environmental Protection
Director-General,
Environmental Protection



Mr Ng Chun Pin
Deputy Chief Executive
Officer, Planning, Corporate &
Technology



Mr Chew Ming Fai
Deputy Chief Executive
Officer, Public Health
Director-General, Public
Health



Ms Koh Li-Na
Assistant Chief Executive,
Meteorological Services and
Radiation Protection
Director-General,
Meteorological Service



Ms Tan Li Yen
Group Director,
Clean Environment



Mr Tan Eng Kim Francis
Group Director,
Corporate Services



Dr Ng Lee Ching
Group Director,
Environmental Health
Institute



Mr Tony Teo
Group Director,
Environmental Public Health
Operations



Mr Andrew Low
Group Director,
Hawker Centres



Mr Desmond Tan
Group Director,
Infrastructure & Engineering



Mr S Satish Appoo
Group Director,
Joint Operations &
Technology



Mr Lim Yui Chien
Group Director,
Public Engagement



Mr Fong Peng Keong
Group Director,
Public Health Services &
Capabilities



Mr Ang Kok Kiat
Group Director,
Radiation Protection &
Nuclear Science



Mr Chan Chin Kai
Group Director, Resource &
Sustainability
Group Director (Infra
Projects), Infrastructure &
Engineering
Chief Sustainability Officer



Ms Ivy Ong Bee Luan
Group Director,
Strategic Planning & Policy



Mr Ng Keng Wei
2 Group Director, Joint
Operations & Technology
Chief Technology Officer



BOARD

The NEA Board⁴ provides strategic direction and oversight of organisational functions and goals, including guiding NEA management in the development and implementation of policies and strategies, to steer NEA in fulfilling our vision and mission. The Board oversees all of NEA's internal and external sustainability efforts, strategies, risks, challenges, and opportunities, including management of NEA's impact on people and the environment. In FY2023, the Board comprised 13 members with diverse and extensive experience from both the private and public sectors.

Led by Chairman⁵ Mr Lee Chuan Seng, the Board has established four committees to execute its responsibilities and ensure high standards of corporate governance. Board members serve on these committees according to their expertise and experience, convening several times a year to assess the risks, challenges, and opportunities within their specific areas of governance. The Board is updated on these discussions at scheduled Board meetings or through email updates.

The Audit Committee reviews NEA's corporate governance and internal control systems, including our annual accounts, audit plans, and audit reports in collaboration with NEA's external and internal auditors. Chaired by Dr Andrew Khoo, it comprises Ms Sylvia Choo (till 31 March 2024), Mr Walter Fernandez (till 31 March 2024), Ms Fang Eu-Lin (from 1 April 2024), Mr Zia Zaman (from 1 April 2024), and Mr Puah Kok Keong.

The Human Resource Committee provides oversight on strengthening NEA's culture, organisation, people, and capability building, and ensures that human resource practices in NEA are objective and equitable. The Committee gives guidance on enhancing leadership effectiveness to maintain a sustainable pipeline of leaders and specialists. Chaired by Mr Lee Chuan Seng, the Human Resource Committee comprises Mr Zia Zaman (till 31 March 2024), Mr Chaly Mah (from 1 April 2024), Ms Teo Siew Pan (from 1 April 2024), and NEA's Chief Executive Officer (CEO).

The Board Finance Committee provides strategic guidance to the management and oversight of investment of surplus funds, cash and working capital management, as well as other financial and fundraising matters. Chaired by Mr Quek Suan Kiat, the Finance Committee comprises Mr Suhaimi Zainul-Abidin (till 31 March 2024), Dr Azlinda Anwar (from 1 April 2024), Ms Janet Young, and NEA's CEO.

Established on 1 April 2023, the Board Risk Committee holds the collective responsibility for the management and reporting of Enterprise Risks to the Board, and provides an overview of the key risks, risk management framework, and the actions taken to mitigate enterprise risks. Chaired by Mr Lim Eng Hwee, it comprises Ms Susan Chong (till 31 March 2024), Ms Fang Eu-Lin (till 31 March 2024), Ms Tan Man Ee (from 1 April 2024), Mr Walter Fernandez (from 1 April 2024), and NEA's CEO.

The NEA Board is bound by a Code of Conduct, comprising five key principles:

- Work with the elected Government to serve the people of Singapore and shape Singapore's future
- Uphold the integrity and reputation of the Public Service
- Ensure there is no conflict of interest between its official duties and personal interests
- Be fair and impartial in carrying out its responsibilities, and not be corrupt or seen to be so
- Exercise prudence in managing public resources

Board members are obligated to disclose their interests through annual declarations. In cases of potential conflicts of interest, they do not participate in the deliberation of transactions or projects with which they may have direct or indirect involvement.

SENIOR MANAGEMENT

Under the guidance of our CEO, our senior management executes NEA's strategic initiatives and advances our organisational excellence.

Members of senior management oversee financial, manpower, and operational aspects of their respective divisions. They also collaborate to improve NEA's performance in risk management, innovation, and sustainability. Regular meetings are convened to review our progress and recalibrate our direction. Strategic projects are commissioned to build NEA's capabilities as both a regulator and champion for the environment.

The compensation package for senior management comprises fixed and variable components. The variable portion, influenced by economic conditions, correlates directly with individual performance. The ratio of variable pay to fixed pay for senior management is relatively higher as this drives performance. Performance is rigorously evaluated and assessed by the Human Resource Committee, which approves corresponding bonuses and salary adjustments.

⁴ All Board members, except NEA's CEO, are non-executive members. Board members are sourced from the Public Sector Division (PSD)'s database, and recommendations by MSE and NEA senior management. The selection of Board members is guided by PSD and the Council for Board Diversity, and takes into account guidelines pertaining to minority groups and female representation on Boards. Board members are selected based on their expertise in enhancing NEA's Board composition, and to complement members who are stepping down or who could contribute to new areas of focus or priorities. Similar considerations are applied when selecting Board members to serve in Board Committees. As NEA Board members are appointed by the Minister for Sustainability and the Environment, shortlisted candidates are proposed to MSE for endorsement as part of the Board renewal process. NEA also aligns with PSD's guidelines on a maximum tenure of six years for public officers on Boards, as well as the maximum number of directorship appointments for Board members on public Boards.

⁵ Chairman is not a senior executive in the Agency



AUDIT

NEA's accounts are audited annually by external auditors, approved by the Board, and published with the Annual & Sustainability Report.

The external auditors are required to declare their state of independence to the Audit Committee prior to their appointment by the Minister for Sustainability and the Environment, in consultation with the Auditor-General. This provides the Board with reasonable assurance that NEA's financial statements are free from material misstatements. The audit is conducted in accordance with the audit plan presented to the Audit Committee, which also approves any significant non-audit services by the external auditors.

Other government agencies serve as external controls to reinforce ethical and responsible conduct within NEA. The Auditor-General's Office and Accountant-General's Department perform scheduled and ad hoc audits to ensure that NEA continues to adhere to good governance practices.

NEA's Internal Audit Division evaluates the effectiveness of governance, risk management, and control processes within NEA. It provides independent, objective assurance and consulting services to improve NEA's operations.

SUSTAINABILITY GOVERNANCE

Sustainability in NEA is governed by a structured framework, with senior management defining the sustainability direction. All NEA Directors are appointed as Sustainability Champions to drive sustainability initiatives within their respective divisions. Additionally, each division appoints an officer as an Eco-office Champion to assist the Director in implementing sustainability initiatives.

These Eco-office Champions play a crucial role in helping NEA achieve our sustainability goals. In addition to collaborating closely with NEA's Sustainability Office to monitor and report the resource consumption performance of their premises, they help to identify potential resource conservation measures, such as reducing electricity usage and water consumption.

SENIOR MANAGEMENT SUPPORT

Senior Leadership Team

To provide overall leadership and direction on sustainability efforts in NEA



MIDDLE MANAGEMENT SUPPORT

Sustainability Committee

To discuss and track environmental sustainability efforts, and review NEA's resource consumption trend

Chair: Group Director, Resource & Sustainability
Committee Members (Sustainability Champions): Directors

Sustainability Champions

To help drive sustainable changes within their divisions, departments, and off-site premises



STAFF INVOLVEMENT

Eco-office Champions

To drive environmental sustainability



● ABOUT THE NATIONAL ENVIRONMENT AGENCY

NEA was established on 1 July 2002 by the NEA Act 2002, as a statutory board under MSE.

Our work is diverse. As a regulator and advocate for the environment, NEA strives to formulate policies that benefit Singapore's environment, economy, and society in the long run.

NEA's strategies steer us towards creating a liveable and sustainable Singapore. We aim to achieve our vision of a clean environment by deepening collaborations, engendering stakeholder support, and strengthening international relations. To drive sustainability initiatives, we regularly partner other public agencies, and engage the industry, non-governmental organisations (NGOs), and the community.

Our upstream value chain includes the Ministry of Finance, MSE, and the Public Service Division (PSD). At the downstream are businesses, hawkers, schools and research institutions, and the general public.

NEA is a member of:

- United Nations Global Compact (UNGC)
- Global Compact Network Singapore, a local chapter of UNGC
- Waste Management and Recycling Association of Singapore Executive Committee
- ASEAN-Japan Energy Efficiency Partnership

We are guided by the NEA Service Charter⁶ that sets out our collective commitment to carry out work with dedication and passion, and provide a trusted and reliable public service for all.⁷

To Ensure a Clean and Sustainable Environment for Singapore



High
Public Health
Standards



Safe, Healthy and
Conducive Living
Environment



A Sustainable and
Resource Efficient
Singapore



Timely and Reliable
Weather and
Climate Services



Vibrant Hawker
Culture

Together with Our Partners and the Community



Environmental Services
Industry as a Force
Multiplier for Mission
Success



Active Community as
Stewards of the
Environment



Public Sector Taking the
Lead in Environmental
Sustainability



International
Partnerships as Strategic
Levers

Enabled by a High Performance and Future-ready NEA



Creation of Capacity



Strong Capabilities for
Mission Success



Strong Corporate Practices for
a Sustainable Organisation



Conducive Culture for
Collaboration, Learning and
Innovation

⁶ NEA's Service Charter can be found at www.nea.gov.sg/corporate-functions/contact-us/service-standards

⁷ For inclusiveness and accessibility, there are various channels for the public to provide feedback and make enquiries, such as the NEA hotline, online feedback form, OneService mobile application, and myENV mobile application. An appeal form is also available on our website to guide the public on filing appeals. The cases are consolidated into NEA's Customer Relationship Management System, which tracks all interactions from the initial report through investigation outcomes to final resolution, and facilitates collaboration with other agencies on inter-agency issues. We have also introduced 'Ask Captain Green', a virtual intelligent chat assistant for the public to obtain instant answers on frequently asked questions

MATERIAL ISSUES

NEA's material issues are reflected in our corporate key performance indicators (KPI), which are approved by our CEO, endorsed by the Board, tracked within NEA, and reviewed every few years based on work priorities and critical areas. We are in the process of refreshing our strategy map and material assessment, and will include them in next year's report.

This table details the impact of our strategies, stakeholders involved, and relevant risks and opportunities, to align with the strategic objectives outlined in NEA's strategy map. These material topics were identified through our strategy mapping exercise.

We work closely with key stakeholder groups to make a positive impact on Singapore's environment.



Public Sector
Agencies



Schools and the
General Public



Employees



Industry, NGOs and
Partners

Material Issue	Impact (Positive and Negative)	Risks and Challenges	Opportunities
High Public Health Standards	Control of vector population; managing public cleanliness, environmental sanitation and hygiene; enforcing smoking-prohibition laws; and providing government after-death facilities and services; are necessary for maintaining the quality of life for Singapore residents 	Vector-borne disease outbreaks	- Use data to understand epidemiology of diseases for temporal and spatial risk stratification - Leverage technology to enhance our operations and build capability in related areas of research
Safe, Healthy and Conducive Living Environment	Monitoring, reducing and preventing environmental pollution; as well as incorporating environmental factors into land use planning, development and building control of new developments; are vital in ensuring a safe, healthy and conducive living environment 	- Deterioration of ambient air quality - Deterioration of coastal and inland water quality	Utilise technology to enhance capabilities in monitoring air and water quality, and identifying pollution sources
Sustainable and Resource Efficient Singapore	Decreasing waste generation and disposal is vital for conserving resources and landfill space. Increasing energy efficiency also helps reduce greenhouse gas (GHG) emissions and combat climate change 	- Resource scarcity - Insufficient waste management capacity - Climate change	- Drive infrastructure enhancement, innovation and behavioural changes to achieve environmental sustainability - Build public trust in our national policy through consistent action

Material Issue	Impact (Positive and Negative)	Risks and Challenges	Opportunities
Timely and Reliable Weather and Climate Services	Timely and reliable meteorological services are vital for informed decision-making and planning 	Severe weather conditions	Boost capabilities in weather monitoring and forecasting through technology
Vibrant Hawker Culture	Keeping hawker culture alive is key to preserving Singapore's heritage and broadening access to economical meals 	Inability to meet growing need for affordable cooked food options in a clean and hygienic environment	- Build new hawker centres and rejuvenate existing ones - Offer a variety of pathways for aspiring hawkers
Private Sector as Force Multipliers for Mission Success	Environmental Services (ES) provide essential cleaning, waste management, and pest management services, to keep the incidences of vector-borne diseases low 	Inability to meet rising demand for ES	Transform the ES industry through the Environmental Services Industry Transformation Map, by helping the industry leverage technology to improve standards and service delivery, build a skilled and productive workforce, as well as grow and export their businesses
Active Community as Stewards of the Environment	As NEA is a frontline agency with multiple public touchpoints, collaboration with stakeholders and public support is critical to mission success 	Lack of environmental ownership	Collaborate with our partners to achieve environmental ownership
Public Sector Taking the Lead in Environmental Sustainability	A Whole-of-Government (WOG) approach is essential to achieving a clean and sustainable Singapore 	- Resource scarcity - Insufficient waste management capacity - Climate change	- Drive innovation and behavioural changes to achieve environmental sustainability - Build public trust in our national policy through consistent action
International Partnerships as Strategic Levers	Cooperation with other countries helps to safeguard and advance Singapore's environmental interests in the global arena 	Fulfilment of NEA's obligations in environmental agreements	Address increasing demand for clean energy, low-emissions transport, and sustainable urban solutions



Material Issue	Impact (Positive and Negative)	Risks and Challenges	Opportunities
Creation of Capacity	Operational capacity is crucial for effective service delivery and enforcement efforts, given NEA's broad spectrum and large number of responsibilities 	Compromised data due to malicious activities against information technology systems and infrastructure	- Deepen technical capabilities in key areas, to improve our services - Simplify application and compliance processes of our regulations
Strong Capabilities for Mission Success	Digitalisation and capability development are key pillars in meeting public sector transformation objectives 	Fulfilment of NEA's manpower requirements	- Uncover new areas for collaboration and deployment of systems and services - Improve productivity for a future-ready NEA
Strong Corporate Practices for a Sustainable Organisation	Financial and manpower management, along with fair and transparent procurement practices, enable effective and efficient operations, governance, and development 	- Growing public expectations - Fraudulent practices attributable to ineffective controls - Talent recruitment and retention - Instances of non-compliance with WSH regulations or procedures	- Promote accountability and resource optimisation - Build public confidence in NEA's financial management as a government-funded organisation - Adopt strong corporate governance and risk management practices - Earn public trust in NEA's responsible use of resources - Develop a stronger sense of identity and purpose among staff - Nurture a more engaged and socially responsible workforce that gives back to the community
Conducive Culture for Collaboration, Learning and Innovation	A strong organisational culture lays the foundation for a high performance and future-ready organisation 	Fragmented information-sharing as teams work in silos	- Enable a high-performance NEA by driving organisational productivity and growth - Strengthen our brand positioning as an employer of choice



ENTERPRISE RISK MANAGEMENT

Our Enterprise Risk Management (ERM) is overseen by the NEA Board, with a dedicated Board Risk Committee.

Robust Risk Management Governance Structure

The Enterprise Risk and Safety Steering Committee (ERSSC), chaired by our CEO, reports to the Board Risk Committee and the NEA Board regularly. The ERSSC ensures that enterprise risks are adequately addressed by providing leadership for our ERM framework, risk assessments, and mitigation plans.

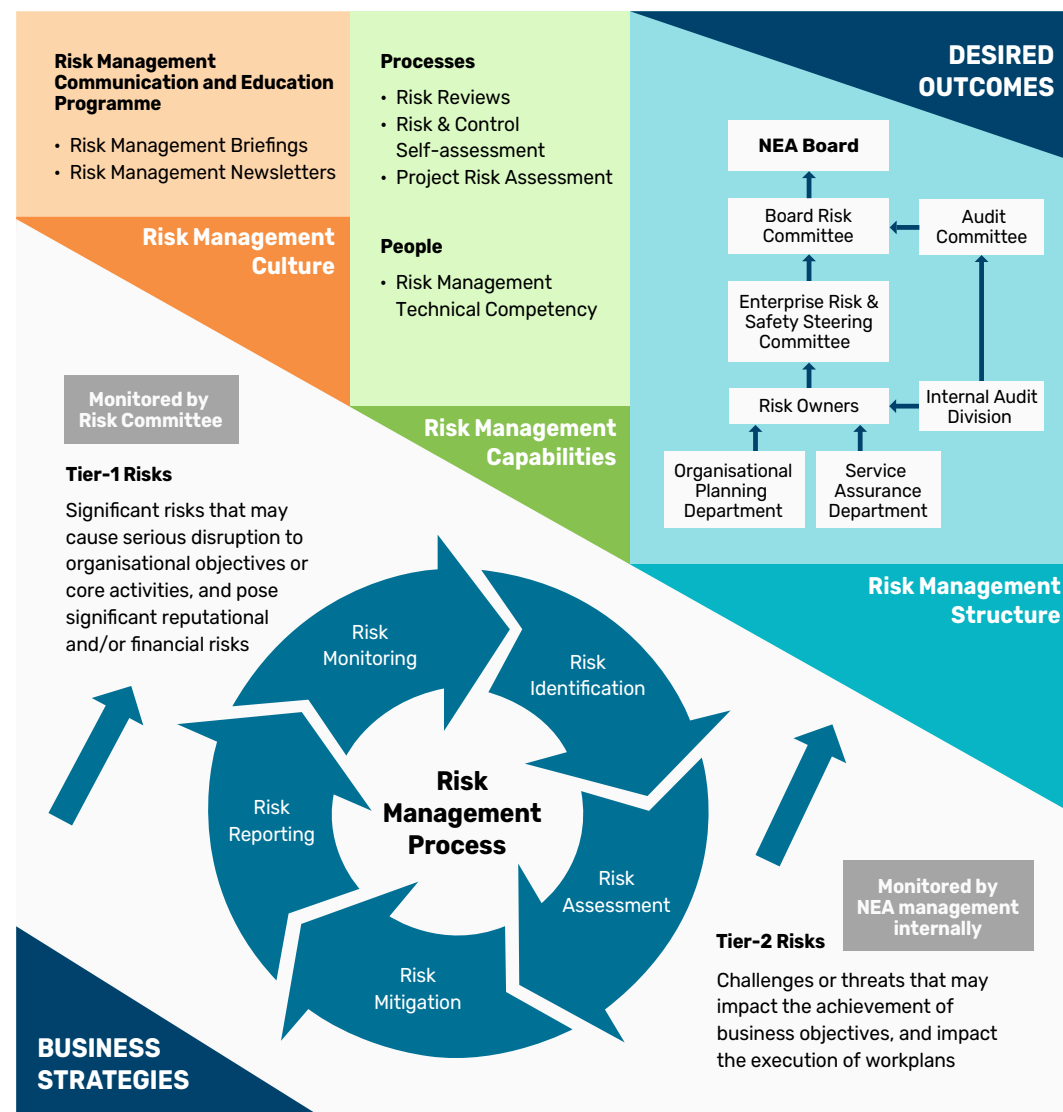
Risk Owners are NEA management tasked with identifying, assessing, and monitoring risks in their respective work areas, and devising treatment plans to mitigate them. These also entail conducting environmental scans on significant issues that could hinder the objectives outlined in the strategy map. Meanwhile, Risk Champions help to drive the risk management efforts in their respective divisions, with Process Owners providing risk-related information, including root causes, consequences and controls, while also implementing and monitoring the risk treatment plans.

Proactive Risk Management Culture

To strengthen our risk culture, and foster greater awareness and understanding of risk management methodologies and tools, NEA undertakes risk communication initiatives that include briefings and newsletters. Staff are encouraged to actively contribute to incorporating risk management practices in their respective areas of work.

Strong Risk Management Capabilities and Tools

NEA continually enhances our risk management capabilities in both our processes and employees. Processes such as risk reviews, project risk assessment, and risk and control self-assessment, are in place to proactively manage risk and control issues. Additionally, structured training programmes equip staff with the necessary knowledge on risk management tools and methodology.



CLIMATE RISKS AND OPPORTUNITIES

NEA is one of the government agencies facilitating efforts to address and mitigate climate change. Our efforts in supporting Singapore's climate resilience and low-carbon transition span across areas such as conducting climate projection studies to model and understand the effects of climate change, promoting energy efficiency in the industry, households and public sectors through legislation, incentives and public education.

In addition, NEA is the administrator of Singapore's carbon tax regime under the Carbon Pricing Act, and has also set out the eligibility criteria under the International Carbon Credit Framework (detailed on page 27). NEA also leads the multi-agency effort to compile Singapore's GHG Inventory, and is a member of the Inter-Ministerial Committee on Climate Change which oversees the WOG coordination on climate change policies to ensure that Singapore is prepared to address climate change.

Nonetheless, as an organisation functioning in a world with climate change, NEA similarly faces risks posed by climate change. As our operations are based in Singapore, NEA is likely to face physical risks that include rising temperatures, heatwaves, rising sea levels, and precipitation, which may have an impact on our operations and maintenance costs. Extreme weather events may also result in disruption of essential services to the public and increased costs for NEA.

Transitional risks that NEA may face include:

- Technology risks, relating to the costs to transition to lower emissions technology and equipment
- Impact of climate-related regulations, policies and carbon prices, which may result in higher operating costs

In the coming years, we aim to progressively understand the materiality of these risks to the organisation.

NEA strives to manage and mitigate such risks and their impact by investing in research, science and technology. We work closely with our partners, and ensure that emerging issues are adequately identified and addressed. We will also strengthen our preparedness and response to mitigate and manage incidents.



UPHOLDING HIGH PUBLIC HEALTH STANDARDS

NEA upholds stringent standards and implements measures in our commitment to safeguard public health.

With the designation of 2024 as the Year of Public Hygiene, NEA is stepping up efforts in tackling cleanliness hotspots with increased enforcement blitzes and surveillance, establishing a Public Toilet Taskforce to address unhygienic public toilets, and stepping up enforcement against poor trade refuse management with the Singapore Food Agency (SFA). These initiatives will bolster vector control and industry capabilities, leverage technology, and mobilise the community to maintain high public health standards in Singapore.

To meet the needs of Singapore's ageing population, NEA is expanding our after-death facilities and services while optimising the use of land. We will also prepare for future outbreaks, taking in insights from the COVID-19 pandemic.

KEEPING OUR COMMON DINING SPACES CLEAN

To promote good hygiene and social responsibility, NEA and SFA have implemented stricter measures against table littering. From 1 June 2023, NEA and SFA stepped up enforcement efforts by issuing written warnings to first-time offenders who failed to return their used trays and crockeries at hawker centres, coffeeshops, and food courts. Repeat offenders were issued fines or charged in court. As at 31 March 2024, approximately 5,000 written warnings and eight fines have been issued.

In April 2024, the average Tray and Crockery Return Rate was 93 per cent, above the 65 per cent in August 2021 before the start of table littering enforcement. These encouraging results would not have been possible if not for our cleaners and the majority of diners, who have contributed to keeping our dining places clean and hygienic.

DETERRING HIGH-RISE LITTERING

The statutory presumption provision for littering from residential flats came into force on 1 July 2023. From this date, if it is proven in any proceedings that littering acts have been committed from a residential flat, the registered owners will be presumed to have committed the offence. If the entire flat is leased, this will apply to the registered tenants of the unit.

This presumption can be rebutted by the owner or tenant by providing proof of absence at the time of the offence, or the identity of a person reasonably believed to be the offender, within 14 days of being required to do so. The first two cases were convicted in June 2024.

AUTOMATING WATERWAY MAINTENANCE

Since early 2023, NEA has been collaborating with cleaning service providers and a robotics solution provider to trial an autonomous solar-powered cleaning boat for removing flotsam and maintaining clean waterways. Successfully completed in January 2024, the trial at Sungei Seletar reduced cleaning manpower required by 80 per cent.

By the end of 2024, NEA plans to deploy additional autonomous cleaning boats in the Central and North-East regions in partnership with the service provider for the two regions.

RAISING THE BAR FOR CLEANING BUSINESSES

The revised Cleaning Business Licensing framework, aimed at driving capabilities and elevating public health standards within cleaning businesses, came into effect on 1 January 2024. These revisions aim to transform the sector by developing its competencies, cultivating a proactive and productive culture, and nurturing a skilled and resilient workforce, thereby improving employment opportunities.

The enhancements will also offer service buyers greater assurance regarding the reliability of cleaning businesses. Service buyers may opt to procure from cleaning businesses with higher classes of cleaning business licences, depending on specific requirements, such as the scale of cleaning works, the utilisation of cleaning equipment and machinery, and the necessary skills and competencies of cleaners.

ADJUSTING THE ORCHARD ROAD NO SMOKING ZONE

Public areas within the Orchard Road precinct have been designated as a No Smoking Zone (NSZ) since 1 January 2019, permitting smoking only in designated smoking areas. Since its implementation, we have been working closely with the Orchard Road Business Association and nearby stakeholders to address feedback on smoking activities in the vicinity.

From 1 February 2024, an adjustment was made to the boundaries of the Orchard Road NSZ along Exeter Road and Killiney Road. Specifically, the area outside Exit A of Somerset MRT station near 111 Somerset was included within the NSZ. To ensure a smooth transition, NEA adopted an advisory approach during the first two months, issuing verbal warnings to individuals caught smoking in the adjusted areas. Fines were issued from 1 April 2024.

TACKLING RAT ISSUES

NEA's surveillance detected an average of 5,200 rat burrows in each two-month inspection cycle last year, an increase of approximately 25 per cent from 2022.

Data collected was analysed to prioritise interventions, such as alerting managers of premises with rat issues to implement rat control measures. These include enhancing refuse management, practising good housekeeping, sealing potential rat entry points, and conducting rat culling, along with regular inspections and checks to monitor the situation. In the Year of Public Hygiene, our focus is on upstream preventive measures to complement downstream rat population control.

To address rat hotspots holistically, NEA and our partner agencies engage the community to improve refuse management in neighbourhoods. This involves coordinating efforts among multiple stakeholders, and enforcing against non-compliant premises owners for creating conditions favourable for rat propagation and harbourage.

NEA will develop a Singapore Standard on performance measurement for pest management services. The performance benchmarks are aimed at enabling a shift towards outcome-based contracting for the pest management sector. In particular, this will encourage pest management companies to focus on and invest in upstream prevention and mitigation in relation to the management of rats and other pests.

We continue to trial the use of surveillance technology. In 2024, NEA conducted a four-month trial on the use of thermal surveillance cameras to pinpoint areas of rat congregation and harbourage and monitor factors contributing to persistent rat activity for targeted intervention and enforcement. The data guided our inspections and follow-ups with stakeholders, and assisted stakeholders in their control efforts by identifying potential causes of rat infestation.

The stepped-up efforts have yielded positive results since the start of this year – the average number of burrows detected from the two-monthly inspection cycle declined to about 2,700, half of what was observed in the preceding year.

SHAVING THE DENGUE PEAK

2023 saw the highest number of dengue cases recorded globally, during which Singapore reported 9,949 dengue cases. Our case numbers were significantly lower than in previous outbreak years of 2019, 2020 and 2022. An outbreak was averted due to intensified efforts. NEA conducted around 813,000 mosquito inspections island-wide, uncovering approximately 19,600 mosquito breeding habitats, largely outside the *Wolbachia*-deployment sites. About 51 per cent of *Aedes* mosquito breeding sites were found in homes, 38 per cent in public areas, five per cent at construction sites, and six per cent in other types of premises. During this period, 12,100 enforcement actions were taken against property owners for mosquito breeding. About 900 fines and 108 Stop Work Orders were issued to construction sites, with 117 contractors charged in court for repeat offences.

i) Expanding *Wolbachia*-*Aedes* deployment

Project *Wolbachia* has helped to curb dengue transmission, especially in release sites, and prevented larger outbreaks. Across all study sites, the *Aedes aegypti* population has been reduced by 80 to 90 per cent after six to 12 months of releases.

Data from 2019 to 2022 indicates that residents in areas with at least one year of releases are up to 77 per cent less likely to contract dengue. Since March 2024, suppression coverage increased by 130,000 households to cover close to 500,000 households. Alongside commercial partners, we will continue to innovate and scale up the deployment of *Wolbachia* technology in a cost-effective and resilient manner, and develop an additional source of *Wolbachia*-*Aedes* mosquito production to increase coverage.

ii) Collaborating with stakeholders

NEA collaborated with the Inter-Agency Dengue Task Force (IADTF) and Town Councils to intensify mosquito control measures and mobilise employees and contractors to reduce dengue transmission. To build capabilities, the second run of a curated training programme was conducted in March 2023.

iii) Launching the National Dengue Prevention Campaign

Given the high weekly baseline of dengue cases since the start of 2023, the National Dengue Prevention campaign was launched early in March 2023 – ahead of the dengue peak season – with the tagline, 'Little but Lethal'. To encourage collective action in reducing

mosquito breeding habitats, the campaign educated and engaged the public through various media platforms as well as NEA volunteers and community partners, including grassroots organisations, NGOs, and schools.

More than 800 outreach activities were conducted, particularly in red dengue clusters and areas with high *Aedes aegypti* populations. Over 600 NEA volunteers were deployed for dengue outreach activities at around 200 locations across Singapore, to educate residents on ways to prevent and protect against dengue. Partners also promoted similar messages through their channels.

iv) Ramping up campaign efforts for the next peak season

In the first quarter of 2024, the weekly number of reported dengue cases remained persistently high, with over 4,800 cases reported. This figure more than doubled compared to the 2,360 cases reported during the same period last year.

Once again, the National Dengue Prevention Campaign 2024 was launched ahead of the dengue peak season on 31 March 2024. It emphasised that dengue prevention is a social responsibility through the refreshed overarching concept of 'Little Acts, Lethal Impact'. Public education and engagement continued through ground outreach efforts led by local grassroots organisations and NEA volunteers, as well as through NGOs, corporate partners and schools.

v) Measuring the true toll of dengue

As national dengue notifications do not capture all infections, they are limited in reflecting the intensity of disease transmission in Singapore. To address this, regular seroprevalence studies, combined with mathematical models, are conducted to estimate the Force of Infection (FOI) – the rate at which susceptible individuals are infected – and provide insights into the actual number of infections.

A 2021 study revealed that 42.7 per cent of Singapore residents had past dengue exposure and immunity to at least one serotype. When combined with data from earlier serosurveys, the FOI estimates showed that the rate of new dengue infections decreased by about 10 times from the 1960s to the 2010s. Despite the high dengue incidence during the outbreak between 2019 and 2020, the force of dengue infection remains low in Singapore. Thus, the resident population remains susceptible to future dengue epidemics. The survey results facilitated dengue risk assessment and communication.

FLUSHING OUT ZIKA WITH WASTEWATER SURVEILLANCE

The Zika virus (ZIKV), primarily transmitted by *Aedes* mosquitoes, can cause severe effects such as congenital Zika syndrome among vulnerable populations. Many ZIKV infections are mild or asymptomatic, meaning infected individuals might not be detected through case surveillance but can still contribute to virus transmission.

In May 2023, after a Zika outbreak was reported along Kovan Road, NEA swiftly implemented wastewater surveillance for ZIKV, leveraging infrastructure and capabilities developed during the COVID-19 pandemic. In February 2024, this method helped to alert residents in Boon Lay Place when persistent signals were detected following a reported Zika case.

NEA pioneered the use of wastewater surveillance to detect ZIKV, which complements clinical and mosquito surveillance in ZIKV transmission hotspots, providing a comprehensive assessment of transmission risks and helping to monitor the effectiveness of control measures.

BUILDING INTERNATIONAL CAPABILITIES FOR DISEASE CONTROL

Since 2011, NEA's Environmental Health Institute (EHI) has been a World Health Organization Collaborating Centre (WHO CC) for Reference and Research on Arboviruses and their Associated Vectors. EHI's redesignation as a WHO CC in February 2023 affirms NEA's global contributions to advancing vector surveillance and control through capacity building and partnerships.

Alongside WHO, the National Centre for Infectious Diseases, and the Ministry of Foreign Affairs, we organised the 7th Singapore International Dengue Workshop in 2023. This event aimed to promote capability building and knowledge exchange in dengue control and management, attracting over 150 international participants both physically and virtually. We also co-hosted training programmes with WHO on wastewater-based epidemiology and with the International Atomic Energy Agency (IAEA) on the sterile insect technique. In addition, EHI actively collaborates with international NGOs and the United Nations (UN) Member States on capacity building and horizon scanning. This includes hosting a regional surveillance network – UNited in Tackling Epidemic Dengue (UNITEDengue) – for cross-border sharing of dengue surveillance.

EHI scientists hold 19 international expert and adjunct academic appointments in various environmental public health domains. These include Chair of the WHO Technical Advisory Group on the Global Integrated Arboviruses Initiative, and member of committee to develop the American Society of Heating, Refrigerating and Air-Conditioning Engineers' new standard on controlling infectious aerosols in built environments. EHI scientists have also been invited to speak at various international platforms, including the Global Consortium for Wastewater and Environmental Surveillance for Public Health.

DETECTING AND INTERCEPTING HAZARDS

Established in 2020, NEA's analytical chemistry laboratory plays an important role in environmental risk assessment for chemical pollutants, including pesticides, volatile organic compounds, and heavy metals. Our field sampling and laboratory analysis capabilities have enabled us to act swiftly to investigate potential environmental contamination cases. In October 2023, we conducted soil tests for pesticides as part of a joint investigation with the National Parks Board (NParks).

DELIVERING AFTER-DEATH FACILITIES AND SERVICES

In 2023, construction commenced for the Mount Vernon Funeral Parlour Complex, which will provide additional wake spaces upon completion. This complements other ongoing development projects, such as the upcoming Mandai North Crematorium slated to be completed in 2025, and a second inland ash scattering garden – the Garden of Serenity. These are developed to meet the anticipated demand for after-death facilities driven by Singapore's changing population demographics.

To better serve the next-of-kin and general public, NEA continues to enhance our service delivery and operational response at our after-death facilities.

SAFEGUARDING SINGAPORE'S ENVIRONMENT

NEA monitors and prevents air and water pollution, and regulates hazardous substances and toxic industrial waste. We ensure a quality living environment through upstream land use planning and siting of industries, and the implementation of noise management measures. We also safeguard the public, radiation workers, and the environment against the harmful effects of radiation. NEA is responsible for pollution control, from front-end development control and licensing work to downstream environmental monitoring and hazardous substance control.

Beyond regulating industries, NEA also takes a proactive and more upstream approach to pollution prevention. Our initiatives include enhancing monitoring and licensing regimes for new pollutants and chemicals of concern, supporting efforts to ensure the secure transport of radioactive materials, and collaborating with agencies across sectors to tackle increasingly complex pollution issues.

In response to the oil spill incident in June 2024, NEA swiftly mobilised teams and coordinated a multi-stakeholder landside cleaning effort. This included joint clean-up operations, air and water monitoring, and volunteer mobilisation. Clean-up of affected beaches have completed in August 2024, a month ahead of schedule.

This year, we have intensified efforts to curb vehicular noise and emissions, made our air and water monitoring results more accessible to the public, and strengthened protocols with neighbouring countries to manage emergency chemical spills. With vigilant oversight of air and water quality and rigorous pollution controls, NEA will continue to play a pivotal role in preserving Singapore's environment.

KEEPING TABS ON AIR QUALITY

NEA monitors six key pollutants across Singapore through an extensive network of ambient air monitoring systems. Hourly air quality updates are available on the NEA website, Haze microsite and myENV app. We will continue to work towards achieving the Long-term Ambient Air Quality Targets.

Air Quality

KPI	2021 Results	2022 Results	2023 Results	Long-term Targets
PM _{2.5} (annual mean)	12.0 µg/m ³	11.0 µg/m ³	11.0 µg/m ³	10.0 µg/m ³
PM _{2.5} (24-hour mean, 99th percentile)	28.0 µg/m ³	26.0 µg/m ³	39.0 µg/m ³	25.0 µg/m ³
PM ₁₀ (annual mean)	28.0 µg/m ³	24.0 µg/m ³	24.0 µg/m ³	20.0 µg/m ³
PM ₁₀ (24-hour mean, 99th percentile)	51.0 µg/m ³	51.0 µg/m ³	62.0 µg/m ³	50.0 µg/m ³
SO ₂ (24-hour mean, max)	89.0 µg/m ³	37.0 µg/m ³	20.0 µg/m ³	20.0 µg/m ³
O ₃ (8-hour mean, max)	176.0 µg/m ³	124.0 µg/m ³	154.0 µg/m ³	100.0 µg/m ³
NO ₂ (annual mean)	25.0 µg/m ³	25.0 µg/m ³	23.0 µg/m ³	40.0 µg/m ³
NO ₂ (1-hour mean, max)	123.0 µg/m ³	128.0 µg/m ³	122.0 µg/m ³	200.0 µg/m ³
CO (8-hour mean, max)	1.2 mg/m ³	1.7 mg/m ³	1.5 mg/m ³	10.0 mg/m ³
CO (1-hour mean, max)	1.3 mg/m ³	2.1 mg/m ³	1.8 mg/m ³	30.0 mg/m ³

GUIDING THE PUBLIC THROUGH TRANSBOUNDARY HAZE

On 7 and 8 October 2023, the air quality in Singapore deteriorated to the 'Unhealthy' range, where the 24-hour Pollutant Standards Index (PSI) exceeded 100 in the East, Central and South regions progressively. The 24-hour PSI peaked at 123 on 7 October 2023 in the East region between 8pm and 9pm.

In response, the Haze Task Force, chaired by NEA and comprising members from 28 agencies, was convened to implement the action plans to mitigate the effects of the transboundary smoke haze. Media releases were issued to inform the public of the multi-agency action plans tailored for various groups, including school children, outdoor workers, and vulnerable individuals. In addition, daily haze advisories were issued from 7 October 2023 onwards, providing the air quality forecast, or 24-hour PSI, for the next day, along with relevant health advisories.

The daily haze advisories ceased on 24 October 2023 when the ASEAN Specialised Meteorological Centre downgraded the risk of transboundary haze occurrence from Level 2 (likelihood of transboundary haze) to Level 1 (dry season).

MOVING TOWARDS CLEANER VEHICLES

The Singapore Green Plan 2030 outlines our national vision of having all vehicles run on cleaner energy by 2040 and our efforts to promote the adoption of electric vehicles.

Commercial Vehicle Emissions Scheme (CVES) and Early Turnover Scheme (ETS) have been extended for two years from 1 April 2023 until 31 March 2025, to encourage the adoption of cleaner commercial vehicles. From 1 April 2023, pollutant thresholds and incentives under the CVES were adjusted to ensure that the total cost of owning and using a cleaner vehicle over its lifespan remains lower than a more pollutive one. The ETS incentive complements CVES to help bridge the higher upfront cost of electric Light Commercial Vehicles, furthering the push for cleaner energy vehicles. The adjustments also took into account the adoption of the Worldwide Harmonised Light Vehicles Test Procedure (WLTP) as the sole test procedure for new light commercial vehicles since 1 April 2023.

To continue encouraging the purchase of cleaner car models, the enhanced Vehicular Emissions Scheme has been extended for three years from 1 January 2023 to 31 December 2025. From 1 January 2024, the VES pollutant thresholds were tightened to make a clearer distinction between pure internal combustion engine cars and cleaner alternatives such as hybrids and electric vehicles. The full switch to the WLTP for passenger cars (including taxis) took effect from 1 January 2024.

CLAMPING DOWN ON ERRANT MOTORISTS

To mitigate vehicle exhaust and noise emissions, all vehicles, including those registered abroad, must adhere to Singapore's exhaust and noise emission standards as outlined in the Environmental Protection and Management (Vehicular Emissions) Regulations. In FY2023, NEA, the Land Transport Authority (LTA), and the Singapore Police Force (SPF) conducted joint enforcement blitzes to reinforce compliance with environmental and road safety regulations. These efforts resulted in 175 enforcement actions against errant motorists for vehicular emissions offences.

MONITORING RECREATIONAL WATER QUALITY

NEA routinely monitors the water quality at seven popular recreational beaches across Singapore, including Changi Beach, East Coast Park Beach, Pasir Ris Beach, Punggol Beach, Seletar Island Beach, Sembawang Park Beach, and Sentosa Island Beaches. At the start of each year, these beaches are graded based on recreational water quality guidelines by WHO, which takes into account the *Enterococcus* (EC) bacteria levels over the past three years.

The grades range from 'Very Good' to 'Very Poor,' with those rated 'Good' or 'Very Good' deemed suitable for primary contact water activities such as swimming.

In the 2024 assessment, the gradings for Pasir Ris Beach and Sembawang Park Beach were downgraded from 'Good' to 'Fair' due to increased occurrences of elevated EC bacteria levels. NEA issued a beach advisory on 5 February 2024 to advise the public against swimming and other primary contact water activities at both beaches. Non-primary contact water activities such as sailing, kayaking, and canoeing could continue as usual. NEA engaged various water sports operators at the two beaches, focusing particularly on the suitability of water activities for recreational waters, advisories for their customers, and adjustments to the water activities offered.

MSE formed an inter-agency committee involving NEA; PUB, Singapore's National Water Agency; NParks; and SFA to investigate and mitigate the potential sources contributing to the elevated EC bacteria levels.

CURBING CONSTRUCTION NOISE

NEA regulates construction noise levels by enforcing specific noise limits outlined in the Environmental Protection and Management (Control of Noise at Construction Sites) Regulations. These limits vary according to time periods and the types of premises; stricter thresholds are imposed during evening and nighttime hours, particularly for noise-sensitive locations like hospitals and residential buildings. Construction sites within 150 metres of these areas are also subject to a no-work rule on Sundays and public holidays.

Since 1 October 2022, contractors violating the no-work rule are not just subject to financial penalties, but also required to install electronic video surveillance systems. These systems allow NEA to monitor construction sites remotely, increasing operational efficiency while deterring future violations. NEA has implemented technological solutions, mandating contractors to install real-time noise meters for continuous monitoring of construction noise levels.

In 2023, NEA received 12,826 construction noise feedback reports. 547 enforcement actions were taken against contractors for violating noise limits and working during prohibited periods in 2023. Approximately 64 per cent of these actions were due to noise limit breaches, primarily involving persistent noise during the day without adequate mitigation measures, and construction work extending into the night.

DEVELOPING EMERGENCY RESPONSES FOR CHEMICAL SPILLS

Under the Malaysia-Singapore Joint Committee on the Environment, NEA has co-developed an emergency response plan with Malaysia's Department of Environment to address accidents involving the transportation of hazardous chemicals on the Tuas Second Link. The plan enables the emergency response agencies of both countries to act swiftly and effectively in order to minimise the consequences of such accidents. Joint field exercises are conducted regularly to familiarise these agencies with the plan and test the programme's effectiveness in coordinating responses between both countries.

The successful completion of the 13th exercise on 26 July 2023 reaffirmed the commitment and preparedness of both countries to respond effectively to potential chemical spill incidents.

As the regulator for radiation protection and nuclear safety, we ensure the safe and secure use of radioactive materials, as well as ionising and non-ionising irradiating devices.

Since Singapore joined the IAEA in 1967 and passed the Radiation Protection Bill in 1973, the application of radiation and use of radioactive materials have expanded, with the most recent development being the opening of the Goh Cheng Liang Proton Therapy Centre in 2023. To ensure a safe environment for all, NEA continues to enhance our competencies to keep pace with advances in technology and applications.

Countries in the ASEAN region and globally are showing renewed interest in nuclear energy technologies as a pathway to achieve their net carbon zero goals. Singapore is also exploring low-carbon alternatives to decarbonise our power sector in the longer term. NEA is strengthening our capabilities to better understand developments in advanced nuclear energy technologies, and to assess their safety and environmental implications. We will also strengthen regional nuclear safety, security, and safeguards through regional and multilateral forums.

REINFORCING SINGAPORE'S COMMITMENT TO NUCLEAR SAFETY

NEA has represented Singapore at the Convention on Nuclear Safety (CNS) since Singapore became a party to the Convention on 15 March 1998. By establishing international benchmarks for adherence, the CNS obligates participating states with land-based nuclear power plants to uphold stringent safety standards. These efforts are advanced through periodic peer reviews where parties meet to evaluate and provide feedback on each other's efforts and measures in ensuring and strengthening nuclear safety.

For parties without nuclear facilities, such as Singapore, this process focuses on emergency preparedness and response, as well as the regulatory framework for radiation protection. Fulfilling our obligations to the CNS, NEA prepared and presented Singapore's ninth CNS national report with inputs from relevant agencies.

In March 2023, NEA represented Singapore at the Joint 8th and 9th CNS Review Meeting, where we delivered a presentation and exchanged insights with other countries on the competencies required for a high level of nuclear safety.

ENHANCING REGIONAL NUCLEAR SAFETY, SECURITY, AND SAFEGUARDS

As Singapore's representative to the ASEAN Network of Regulatory Bodies on Atomic Energy (ASEANTOM), NEA attended the 10th ASEANTOM Annual Meeting in Indonesia from 23 to 24 July 2023. During the meeting, members exchanged information and updates on activities by the five ASEANTOM Technical Working Groups, and explored opportunities for future collaborations, including with international partners.

In December 2023, the ASEANTOM Technical Working Group on Hazard Assessment and Radiological Dispersion Modelling commenced a 30-month ASEAN-European Commission project, titled 'Enhancing and Strengthening the Response to a Radiological or Nuclear Emergency in ASEAN – Decision Support System (DSS)'. Jointly developed by Singapore and the European Commission for the 10 member states, it encompasses training sessions, tabletop exercises, and further customisation of the DSS for ASEAN's context. These aim to enhance capabilities in utilising the DSS and responding to emergencies, ultimately promoting coherence within ASEAN.

Through the ASEANTOM Technical Working Group on Radiation Monitoring which Singapore co-leads, ASEAN also partnered the European Commission to enhance the region's radiation monitoring capability with an additional 82 monitoring stations to be set up in six ASEAN Member States. A data exchange arrangement and platform would also be developed to facilitate the sharing of radiation data by all ASEAN Member States. As at end 2023, about a third of the additional monitoring stations have been set up, commissioned, and are transmitting data to the data sharing platform.

NEA actively collaborates with international partners to enhance the ASEAN Member States' understanding of international standards and best practices in nuclear safety, security, and safeguards. NEA hosted a regional seminar from 5 to 9 June 2023 on the institutionalisation of safeguards and innovations in safeguards implementation, in collaboration with the National Nuclear Security Administration under the United States Department of Energy. The seminar sought to identify attributes for sustainable nuclear safeguards implementation, develop approaches to sustain safeguards during disruption, and facilitate the sharing of insights among countries.

From 11 to 14 November 2023, NEA also hosted a regional workshop on conducting transport security inspections for Asian countries, in collaboration with IAEA. The workshop aimed to reinforce the importance of robust security inspection programmes when transporting nuclear and radioactive materials and build capacity for executing them.

SHOWCASING RADIATION EXPERTISE ON THE INTERNATIONAL STAGE

NEA continues to strengthen our international standing by participating in IAEA expert missions. From 26 November 2023 to 6 December 2023, an NEA officer was a reviewer during the IAEA Integrated Regulatory Review Service Mission to Morocco, which aimed to enhance the effectiveness of national regulatory infrastructure concerning nuclear, radiation, radioactive waste, and transport safety.

Two NEA officers were invited as a reviewer and an observer during an IAEA Occupational Radiation Protection Appraisal Service (ORPAS) Mission to Thailand from 10 to 19 March 2024. The ORPAS mission involved evaluating legislation, regulations, and the practical implementation of requirements related to occupational radiation protection at facilities and technical service providers.

NEA officers were also called upon as technical experts to deliver a lecture on 'Basic Gamma Spectrometry Techniques' at an IAEA Regional Workshop on 'In-situ Radiation Detection Demonstration and Basic Gamma Spectrometry Techniques for Enhancing Emergency Preparedness and Response Capabilities', held virtually on 20 and 21 February 2024.

ADVANCING RADIATION CONTROL WITH TECHNOLOGY

NEA leveraged artificial intelligence and robotic process automation to introduce an e-commerce surveillance tool in May 2023. The tool helps to manage the online sales of controlled non-ionising irradiating apparatus and unsafe devices, improving our operational efficiency.

In October 2023, we completed a study to assess the radiofrequency radiation levels in public areas of Singapore attributable to radiofrequency radiation emissions from communication and broadcasting services. This data collected from our study proved instrumental to develop better informed policies for controlling non-ionising radiation.

MAINTAINING EXCELLENCE IN INTERNATIONAL RADIOCHEMISTRY

Throughout 2023, NEA's National Radiochemistry Laboratory and Secondary Standards Dosimetry Laboratory maintained exceptional performance in the IAEA international proficiency tests. These tests assess laboratories for their levels of accuracy, and precision within specified acceptance limits. The results validate our technical competencies, and ability to uphold high standards and produce reliable results.

NURTURING A SUSTAINABLE SINGAPORE

Nurturing a sustainable Singapore for future generations is a collective endeavour that requires adopting environmentally friendly practices and responsible resource stewardship. In alignment with these principles, NEA's initiatives help Singapore address global challenges, such as rising energy consumption, carbon emissions, and waste generation.

To curb increasing energy consumption, we focus on promoting sustainable energy use in households and industries by encouraging the adoption of energy-efficient appliances and equipment. Our efforts to reduce waste generation and embrace a resource circularity approach also contribute to mitigating the carbon emissions associated with waste treatment.

STRIVING FOR ZERO WASTE

Under Singapore's Zero Waste Masterplan and the Singapore Green Plan 2030, our goals are to boost the national recycling rate to 70 per cent (80 per cent non-domestic recycling and 30 per cent domestic recycling) and cut per capita waste-to-landfill by 20 per cent by 2026 and 30 per cent by 2030. Meeting these targets will extend the lifespan of Singapore's only landfill beyond 2035.

Over the past decade, domestic waste generated per capita and non-domestic waste generated per dollar Gross Domestic Product (GDP) have decreased by more than 15 per cent and 30 per cent respectively. This was made possible through NEA's numerous waste reduction initiatives, such as the implementation of a disposable carrier bag charge to nudge consumers to bring their own reusable bags and reduce reliance on disposables. Effective from 3 July 2023, operators of SFA-licensed supermarkets with an annual turnover exceeding \$100 million were required to charge a minimum of five cents for each disposable carrier bag provided to customers at their physical stores. NEA also worked with partners across various sectors through the annual 'Say Yes to Waste Less' campaign, to encourage the public to make eco-friendly choices in their daily lives by reducing the use of disposables and food wastage.

The impact of such behavioural nudges was evident in NEA's 2023 survey, which revealed that 72 per cent of households recycled as compared to 64 per cent in 2021. However, the overall recycling rate decreased from 57 per cent in 2022 to 52 per cent in 2023, largely due to the amount of construction and demolition waste – which is almost completely recycled

– dropping significantly last year. The recycling rate for other waste streams, such as paper, declined from over 50 per cent a decade ago to 31 per cent in 2023. This can be attributed to structural factors such as freight costs and commodity prices.

To keep these materials in the resource loop for as long as possible, NEA has developed a range of initiatives that include food waste segregation and the upcoming Beverage Container Return Scheme, which encourages consumers to return their empty beverage containers for recycling. These efforts help alleviate the strain on resources and divert waste from our landfill.

Waste and Recycling Targets

KPI	2013 Results	2018 Results	2021 Results	2022 Results	2023 Results	Targets
Overall recycling rate	62%	62%	55%	57%	52%	70% by 2030
Domestic recycling rate	20%	22%	13%	12%	12%	30% by 2030
Non-domestic recycling rate	77%	76%	70%	72%	67%	80% by 2030
Domestic sector – waste generated per capita	1.08 kg a day/person	0.97 kg a day/person	0.92 kg a day/person	0.90 kg a day/person	0.88 kg a day/person	–
Non-domestic sector – waste generated per \$bn GDP	39.6 tonnes a day/GDP (\$bn)	33.3 tonnes a day/GDP (\$bn)	27.7 tonnes a day/GDP (\$bn)	28.8 tonnes a day/GDP (\$bn)	25.7 tonnes a day/GDP (\$bn)	–

INVESTING IN WASTE AND RESOURCE RESEARCH AND INNOVATION

To ensure the relevance of NEA's investments in research and development efforts under the Closing the Resource Loop Funding Initiative, we conducted a mid-term review (MTR) of our waste and resource management research programmes supported by the Research, Innovation and Enterprise funds. This review assessed the progress and outcomes of funded programmes, identified opportunities for refinement and alignment with Singapore's sustainability goals and challenges, and reprioritised NEA's research and innovation focus areas for continuous support.

The final MTR report, submitted to the National Research Foundation, Singapore in September 2023, outlined key waste and resource management research areas that NEA will continue to pursue. These include building long-term capabilities for residue management, particularly for toxic industrial waste (TIW) and treated TIW; resource circularity; and the recovery of value from priority waste streams such as food, plastic, and e-waste.

Furthermore, NEA will strengthen support for the efforts of the Environmental Services Industry Transformation Map, aimed at enhancing the productivity of companies, service delivery, and recycling rates through automation and digital solutions.

EMPOWERING ZERO WASTE CHAMPIONS IN WORKPLACES

In partnership with NTUC LearningHub, Republic Polytechnic, and the Waste Management and Recycling Association of Singapore, NEA introduced the Zero Waste Manager course to support businesses in building waste management competencies and integrating the 3Rs (Reduce, Reuse and Recycle) into their organisational cultures.

Launched in March 2024, the course encourages companies, especially those covered under the Mandatory Waste Reporting scheme, to formalise waste reduction responsibilities by appointing and training Zero Waste Managers. By equipping them with knowledge and skills to champion waste management in their organisations, the course empowers Zero Waste Managers to help their employers increase resource efficiency, lower costs, and advance corporate social responsibility efforts.

In line with NEA's Environmental Services workforce transformation goals, companies can also leverage Workforce Singapore's Career Conversion Programme for Sustainability Professionals to support individuals in transitioning to sustainability-related roles.

ENCOURAGING RECYCLING IN HOUSEHOLDS

The 'Recycle Right' campaign was kickstarted to encourage households to recycle more and recycle right, and reduce contamination rates in blue recycling bins and chutes by enhancing public awareness of recyclable items. As part of this campaign, NEA launched the Bloobox initiative, urging households to set up their own recycling corners and make recycling a way of life.

Following a successful pilot phase from November to December 2022, nationwide distribution of Blooboxes commenced on 19 March 2023, and over 565,000 Blooboxes were issued. NEA also partnered with various organisations to distribute Blooboxes at community events and engaged primary and secondary schools to promote recycling habits both in school and at home.

As the nation's carbon regulator, NEA advances Singapore's climate change strategy by administering the carbon tax and building capabilities to regulate international carbon credits. While Singapore works actively towards its goal of net zero emissions by 2050, we contribute to our international climate commitments through accurate and timely monitoring, reporting, and verification of national emissions.

Moving forward, NEA aims to expand regulatory expertise in Carbon Capture, Utilisation, and Storage, and to further foster a sustainability-conscious society and economy.

PROMOTING GREEN LIVING WITH CLIMATE VOUCHERS

NEA announced in March 2024 that the Climate Friendly Households Programme would be enhanced to allow all Housing & Development Board households to claim Climate Vouchers worth \$300 from 15 April 2024 onwards. Additionally, the range of energy- and water-efficient appliances and fittings eligible for these vouchers has been expanded to include direct current fans, washing machines, and water closets. Valid until 31 December 2027, these vouchers aim to promote resource efficiency and sustainable living among households while offering long-term utility savings.

Through the Mandatory Energy Labelling Scheme (MELS) and Minimum Energy Performance Standards (MEPS), NEA has helped households save on energy bills and contribute to Singapore's decarbonisation efforts. MELS allows consumers to make more informed purchasing decisions through the use of energy labels. MEPS removes the least energy-efficient appliances from the market entirely. Collectively, they help consumers from being locked into the high energy costs of operating energy-inefficient appliances. In addition to air-conditioners, refrigerators, dryers, televisions and lightings, MELS and MEPS were extended to portable air-conditioners in 2024, with plans to be further extended to water heaters in 2025.

UPHOLDING RIGOUR IN NATIONAL GREENHOUSE GAS REPORTING

The agencies involved in the compilation of Singapore's GHG inventory are committed to refining the GHG inventory to align with principles of the Enhanced Transparency Framework (ETF) for reporting to the United Nations Framework Convention on Climate Change (UNFCCC). The ETF's principles are transparency, accuracy, completeness, consistency, and comparability. This commitment is especially crucial given Singapore's obligation under the Paris Agreement to submit the First Biennial Transparency Report by December 2024.

In order to validate the alignment with ETF principles, NEA worked with UNFCCC to conduct an in-country quality assurance workshop for Singapore's GHG Inventory. Co-hosted by four UNFCCC GHG experts from 11 to 15 September 2023 in Singapore, the workshop also helped expand our capacity to meet the new and more rigorous requirements of the ETF. Approximately 40 attendees from various data owner agencies worked closely with UNFCCC experts to identify areas of improvement. These include the potential development of country-specific emission factors for streams with elevated emissions.

IMPLEMENTING STRINGENT STANDARDS FOR CARBON OFFSETS

NEA and MSE have outlined the Eligibility Criteria within Singapore's International Carbon Credit (ICC) Framework. Effective from 1 January 2024, carbon tax-liable companies can utilise eligible ICCs to offset up to five per cent of their taxable emissions.

The Eligibility Criteria requires ICCs to meet seven internationally recognised principles⁸ to demonstrate high environmental integrity, and represent emissions reductions or removals that occur within the timeframe specified under Article 6 of the Paris Agreement. These principles draw from rigorous international standards, such as the Carbon Offsetting and Reduction Scheme for International Aviation, and were developed in consultation with over 70 stakeholders that spanned industry and private organisations.

Moving forward, NEA will continue to keep abreast of developments in carbon market governance and review the Eligibility Criteria periodically to ensure they align with the highest standards of environmental integrity.

⁸ Details on the ICC eligibility can be found at www.carbonmarkets-cooperation.gov.sg/environmental-integrity/eligibility-criteria

DELIVERING TIMELY AND RELIABLE WEATHER AND CLIMATE SERVICES

In the face of changing weather patterns, we remain steadfast in helping key economic sectors and members of the public plan ahead and prepare. NEA's Meteorological Service Singapore (MSS) continues to spearhead research efforts and enhance long-term projections on climate impact, which are essential for adapting to the evolving realities of climate change.

ADVANCING HEAT STRESS PRECAUTIONS

In collaboration with MSE and supported by the Ministry of Health, we launched the Heat Stress Advisory in July 2023 to help the public make more informed decisions on undertaking prolonged outdoor activities, in order to minimise the risk of heat stress and heat-related illnesses.

Structured around three levels of heat stress risk – low, moderate, and high – the advisory is based on the Wet Bulb Globe Temperature (WBGT), an internationally recognised measurement that reflects the main environmental factors contributing to heat stress, such as air temperature, humidity, wind, and solar radiation. NEA operates an islandwide network of nine WBGT stations, with plans to significantly expand the number of such stations in the coming years.

STRENGTHENING REGIONAL METEOROLOGICAL CAPABILITIES

The ASEAN Specialised Meteorological Centre (ASMC), hosted by MSS, successfully concluded its inaugural ASMC Attachment Programme for officials from ASEAN Member States.

During the northern ASEAN dry season from March to May 2023, one official each from Myanmar and Lao PDR were attached to the ASMC Operations Centre. They actively participated in preparing daily weather and haze assessments and outlooks for the ASEAN region, enhancing their capabilities in regional weather analysis and prediction.

In 2024, the ASMC attachment programme will continue to foster operational capability building and collaborative research projects with officials from ASEAN Member States, further benefiting the region.

UNVEILING NEW CLIMATE INSIGHTS FOR THE REGION

In January 2024, NEA's Centre for Climate Research Singapore (CCRS) released the findings of Singapore's Third National Climate Change Study (V3) projecting higher temperatures, more pronounced wet and dry extremes, and an accelerating increase in mean sea levels for Singapore and Southeast Asia by the end of the century.

These findings are consistent with the Intergovernmental Panel on Climate Change (IPCC) 6th Assessment Report (AR6). V3 downscales global climate models from IPCC AR6 to higher resolution projections for Singapore and Southeast Asia using CCRS' customised Regional Climate Model. Compared to the previous study in 2015 (V2), it offers more granular projections of our future regional climate, leveraging global improvements in climate science research.



• CHERISHING OUR VIBRANT HAWKER CULTURE

Hawker centres and markets are integral social, economic, and cultural institutions in Singapore. Beyond being icons of our culinary heritage, they serve as social spaces that bond the community and provide affordable options for cooked food and fresh produce. As their custodian, NEA ensures that these time-honoured traditions continue to thrive while enriching our collective identity.

As the landscape evolves, NEA will continue to rethink how to better build, operate, and future-proof this cherished national institution.

PROVIDING AFFORDABLE FOOD OPTIONS

Hawkers balance pricing their food affordably and generating a reasonable income. To help hawkers with this, NEA moderates the rent of hawker stalls by not setting reserve rents for stall tenders and by disallowing subletting. NEA continues to explore measures to address increasing operating costs, such as bulk purchasing services and improving hawkers' productivity.

SUSTAINING THE HAWKER TRADE

The hawker trade is evolving, with a new generation of aspiring hawkers joining the trade.

To equip and enable aspiring hawkers to enter the hawker trade, NEA runs programmes that offer reduced stall rental at pre-fitted stalls, apprenticeship opportunities, and equipping skills required to run a hawker business. NEA also continually reviews our policies to make the hawker trade more attractive, such as looking into measures to improve work conditions and career prospects for hawkers.

In addition, NEA helps hawkers to reduce reliance on manpower through various initiatives targeted at improving hawkers' productivity. These programmes were expanded in FY2023. The Productive Hawker Centres programme was broadened to include co-funding for adoption of Cleaning Process Automation solutions, while the Hawkers' Productivity Grant was expanded to include stall-level digital services solutions such as e-ordering. The co-funding amount for the grant was also increased from \$5,000 to \$7,000.

ENHANCING INFRASTRUCTURE FOR OUR COMMUNITY DINING ROOMS

NEA is on track towards building 20 new hawker centres. In FY2023, we opened Buangkok Hawker Centre and reopened a refreshed Jurong West Hawker Centre. These new hawker centres feature high ceilings for good natural ventilation and spacious seating areas, providing patrons with a more comfortable dining environment. Three more will open in 2024 – Anchorvale Village Hawker Centre, Woodleigh Village Hawker Centre, and Punggol Coast Hawker Centre.

Existing hawker centres also underwent infrastructural improvements. Repairs and Redecoration (R&R) was carried out in 14 existing hawker centres in FY2023. In 2024, Geylang Serai Market and Cheng San Market and Cooked Food Centre will undergo the Hawker Centres Transformation Programme, which goes beyond the scope of R&R work to better future-proof its infrastructure.

DOCUMENTING OUR HAWKER CULTURE

In 2023, NEA and the National Heritage Board supported Professor Lily Kong's publication of the second edition of 'Singapore Hawker Centres: People, Places, and Food'.

The 2007 edition captured the socio-economic and cultural progression of hawker centres and street food culture in Singapore. Acknowledging their evolution over the past 16 years, especially with increased government efforts in constructing new hawker centres, the updated edition commemorates hawker culture as Singapore's first element inscribed on the UNESCO Representative List of the Intangible Cultural Heritage of Humanity. It also serves as a valuable resource for both present and future generations.

LIVING OUT EARTH AS ONE NEA

NEA is transforming and making continuous improvements to our areas of work, especially in the areas of cleanliness, waste management operations, after-death, enforcement, and regulations. We are doing so by making the most of technology, bringing out the best in our workforce, and ensuring their safety and well-being.

CEMENTING EARTH VALUES IN NEA'S CULTURE

COVID-19 prompted an evaluation of NEA's values, to prepare our organisation for complexities on the horizon and foster a unified identity under the banner of One NEA.

This initiative involved dialogues and focus group discussions with staff across NEA, to refine our existing values and co-create the EARTH values – empathy, agility, resilience, teamwork, and heart for the environment. The acronym encapsulates the virtues NEA staff uphold in their endeavours to safeguard the environment, serving as a constant reminder of our vision, mission and roles. HEART, an anagram for EARTH, also underscores NEA's dedication to our work.

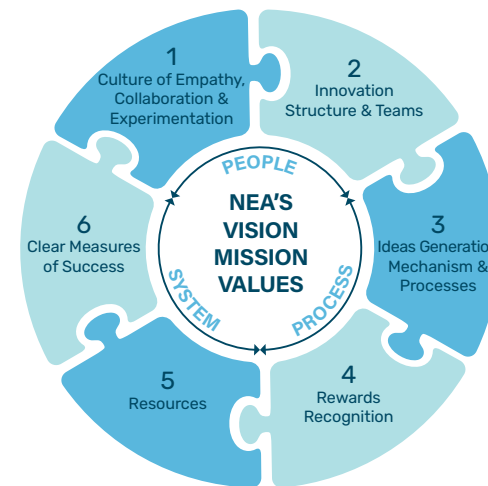
This two-year journey culminated in the launch of EARTH at the NEA Staff Seminar in May 2023, after which we regularly engaged staff to enhance awareness, initiate discussions, and reinforce the EARTH values. Sustaining these efforts ensured that NEA's values would remain top-of-mind and deeply ingrained in our organisational and personal practices.



PURSUING EXCELLENCE AND INNOVATION TO CREATE CAPACITY

To champion an innovation-driven culture, NEA has an innovation framework rooted in organisational development and behavioural insights. It drives excellence and value creation on six different fronts. Our hard work was recognised at the 48th International Convention on Quality Control Circles⁹ in 2023, where we garnered three Gold awards for:

- Achieving standardisation in biological development through engineering precision in Project *Wolbachia*
- Revolutionising surveillance of smoking offences in smoking-prohibited areas
- Smart flushing system for buoy-based sensors



In November 2023, NEA's mobile application, myENV, received the Special Award for Smart Cities - Best in Citizen Well-being at the International Data Corporation Future Enterprise Awards¹⁰ in both Singapore and the Asia Pacific region. This achievement underscores myENV's success in promoting citizen well-being, and its adaptability in the ever-evolving digital landscape.

Under the Environmental Robotics Programme¹¹, "Panthera", an autonomous pavement-sweeping robot created by the Singapore University of Technology and Design with support from the National Robotics Programme office, successfully passed LTA's T1 test for Autonomous Vehicles on Public Paths in May 2023. Moving forward, outdoor trials at Jurong Lake Gardens will be conducted to refine the robot's capabilities. NEA will also explore additional applications for improved productivity and work outcomes.

⁹ The International Convention on Quality Control Circles 2023, themed 'Innovation and Vitality, Lighting up the Beauty of Quality,' marked NEA's 12th year of participation. The annual event showcases the most innovative practices across the globe, with a spotlight on Asia

¹⁰ The International Data Corporation Future Enterprise Awards recognise organisations leading the charge in digital-centric markets and effectively implementing enterprise-level digital-first strategies

¹¹ Launched alongside the Environmental Services Industry Transformation Map, the Environmental Robotics Programme is a cornerstone of NEA's strategy to harness robotics technology and create high-value jobs. This initiative focuses on developing robotic solutions for public space cleaning, waste management, inspection, monitoring, and sampling

We commissioned the Public Cleanliness module within Integrated Operations Management System (IOMS)¹² on 30 September 2023. Predictive models for the module have been developed based on planning parameters, integrating external system data from both NEA and other government agencies. While these models are currently undergoing refinement, the launch of the module has heralded a new concept of operations, phasing out manual planning. As of 27 March 2024, we have used IOMS to successfully generate four rounds of automated monthly audit plans. Looking ahead, we are working in phases to onboard other areas such as Vector Control, to support their operation planning.

Internally, innovation efforts are recognised through NEA and Public Service level awards and initiatives. In FY2023, NEA surpassed our goal for the inaugural NEA 100K Hour Challenge by completing 127 projects involving 392 officers, saving 106,201 hours through more efficient work processes. Officers who participated in the challenge were rewarded a day off. NEA was also among the top three agencies in PSD's Million Hours Challenge, with our strong ground-up innovative culture of 118 innovation projects.

FOSTERING INTERNATIONAL PARTNERSHIPS AND COLLABORATIONS

In 2023, NEA actively engaged our foreign counterparts on environmental protection and sustainability issues on various bilateral platforms. NEA also hosted delegations from various countries on ad hoc visits and study trips on a wide variety of environmental topics, including the Project *Wolbachia*, hawker centre management, waste management and the circular economy, environmental protection, public health, and more.

In addition, NEA signed a Memorandum of Understanding (MOU) with the United States Environmental Protection Agency to reinforce bilateral cooperation in environmental management and protection. We also signed another MOU with the Republic of Korea's National Institute of Environmental Research to share satellite observations on air quality and collaborate on research on its applications for addressing environmental and pollution issues such as climate change and transboundary air pollution.

Within the ASEAN region, Singapore assumed the Chairmanship of the ASEAN Agreement on Transboundary Haze Pollution (AATHP) in 2023. As part of our chairmanship, MSE and NEA hosted the Conference of the Parties to the AATHP as well as the Sub-Regional Ministerial Steering Committee meetings in Singapore. In furtherance of the region's preparedness to address haze, we also worked with Indonesia to reach an agreement on recognising the functions of the Singapore-hosted ASEAN Specialised Meteorological Centre and its co-existence with the ASEAN Coordinating Centre on Transboundary Haze Pollution Control.

NEA actively supported MSE in the ongoing negotiations on the Internationally Legally Binding Instrument on Plastic Pollution, including in the marine environment. This important treaty seeks to galvanise action to address plastic pollution. While Singapore supports ambitious global action, the positions we have advanced in the negotiations aim to ensure that the instrument is pragmatic, implementable, and facilitates effective action by countries to manage plastic pollution.

ENGAGING OUR PARTNERS AND COMMUNITY

The Environmental Services (ES) Awards, a key initiative of the Environmental Services Industry Transformation Map, celebrates the commitment and excellence of the ES workforce through two award categories – the ES Star Award and the ES Achievement Award. At the Clean & Green Singapore Day on 4 November 2023, 27 individuals received the ES Star Award. Meanwhile, another 352 ES employees were honoured with the ES Achievement Award, which was presented at an appreciation event organised by the Environmental Management Association of Singapore on 24 January 2024.

NEA will continue to improve and expand the accessibility and inclusiveness of our feedback channels. These include the NEA hotline, online feedback form, and the myENV mobile application. Additionally, an appeal form is available on NEA's website to guide the public on how to file appeals, making the process more straightforward and transparent. All cases are consolidated into NEA's Customer Relationship Management System which provides a comprehensive view of each case, allowing for more efficient tracking of interactions from the initial report through investigation outcomes to final resolution, ensuring a more coordinated and effective response to the public. The system also facilitates collaboration with other agencies on interagency issues.

NEA believes in developing an engaged and socially responsible workforce that makes a positive impact on the environment and community and contributes to NEA's social sustainability goals. In FY2023, \$59,198 was raised through philanthropic initiatives supported by NEA, excluding donations made through the monthly Community Chest SHARE¹³ programme. Through these, NEA aims to build stronger relationships with our stakeholders and enhance public trust.

¹² The IOMS was developed to streamline operations planning and management across NEA departments, fostering synergy and complementing NEA's existing Integrated Field Operations System

¹³ The Community Chest is the philanthropy and engagement arm of the National Council of Social Service. For details on SHARE, visit www.comchest.gov.sg/brand/share

EMBARKING ON OPERATIONAL REVIEWS

NEA has embarked on five operational reviews to accelerate transformation, create better outcomes, and improve service delivery.

The Waste Management Facilities (WMF) Operational Review aims to enhance safety leadership and culture, improve our safety systems, and inculcate a safety-first mindset in our integrated workforce, while pushing for continuous improvement, supported by technology as an enabler. As part of the review, an External Advisory Panel (EAP) was appointed in November 2023, comprising senior industry leaders experienced in managing large facilities and operations. The EAP provides a strategic external perspective on how operational excellence, total WSH, and resilience at our WMFs can be enhanced, including how systems and processes can be improved. In April 2024, NEA also formed a full-time Safety and Operational Excellence Office within the Waste and Infrastructure Operations Management Division, to drive the safety culture in WMF operations and support the operational WMF teams in enhancing safety of work processes.

The After-Death Services Operational Review focuses on service assurance, building supply resilience, and ensuring a safe and clean environment at the facilities. Key initiatives include leveraging technology to improve back-end processes at Mandai Crematorium, and enhancing safe work procedures for Choa Chu Kang Cemetery's burial operations.

Beyond the Year of Public Hygiene, the Public Cleanliness Operational Review aims to continue the momentum to enhance cleanliness in Singapore. NEA will optimise the cleanliness value chain by integrating automation and cleaning technologies to improve productivity, increasing the use of video analytics and surveillance for enforcement. The review also looks to foster greater stakeholder ownership for improved refuse and rat management.

To strengthen regulatory agility, the Operational Review looks to review our policy approach to better manage environmental issues, streamline our development control process, and explore innovative ways to meet public expectations while managing an increasing workload.

The Dengue Enforcement Administration Operational Review aims to improve NEA's operational effectiveness by streamlining processes and leveraging digital technology to enhance the enforcement process while reducing administrative burdens. This allows NEA to pivot manpower resources from dengue, to address other pressing public health needs when required.

NEA is committed to maintaining this momentum of change, by fostering a culture of innovation and continuous improvement.

PRIORITISING WORKPLACE SAFETY AND HEALTH

WSH is our top priority, and is embedded in our EARTH values.

Endorsed by our CEO, our WSH Policy underscores how safety is an integral part of our organisational culture. Everyone plays an active role in building a positive safety culture, and ensuring a safe work environment for all our staff and stakeholders.

Our WSH efforts are overseen by the Enterprise Risk and Safety Steering Committee, which reports to the Board Risk Committee and the NEA Board. To strengthen WSH oversight and a greater sense of ownership, we established Group WSH committees¹⁴, which are chaired by respective Group Directors. To sustain a robust WSH management system, NEA's WSH team¹⁵ was expanded to become the WSH Division to strengthen our capabilities and capacity to support the WSH leads in the various Groups and Divisions. Additionally, we also strengthened WSH capacity in key operational divisions, including a full-time Safety and Operational Excellence Office within the Waste and Infrastructure Operations Management Division.

We achieved our target of zero fatal injuries and zero major injuries in 2023. We continue to aim for zero fatal injuries and zero major injuries.

NEA's WSH Performance for 2023¹⁶

	No. of Injuries
No. of Fatal Injuries	0
No. of Major Injuries	0
No. of Minor Injuries	26
Workplace Injury Rate (per 100,000 workers)	709

NEA and two of our officers have been charged under the Workplace Safety and Health Act in relation to the Tuas Incineration Plant incident in 2021. The cases are presently before the Courts.

¹⁴ NEA's WSH Committees are organised by premises type and consist of employees from the respective divisions. These WSH Committees inspect workplace accidents, conduct investigations, promote safety, share WSH guidelines, and ensure sufficient safety training is conducted for their divisions. The WSH Committees meet monthly, quarterly or annually depending on their risk tiering, to review WSH matters

¹⁵ Decentralised WSH Teams are made up of Group-level WSH Committees, WSH Representatives, and Safety Officers

¹⁶ WSH statistics are based on the calendar year, so as to allow benchmarking with MOM's national WSH statistics. There were no major injuries for 2023 – major injuries definition is aligned to MOM's terminology, as reflected in the national statistics report

As part of our continuous efforts to strengthen WSH culture, we rolled out education, promotion and engagement initiatives this year. Our inaugural NEA WSH Day was held in September 2023. A basic WSH e-learning course was also developed in-house and launched in 2023, comprising three bite-sized modules to enhance the WSH fundamentals. All employees have completed this e-learning. In March 2024, we rolled out an Open Reporting Campaign to encourage staff to speak up for safety and share their WSH feedback, as well as to actively report any hazards or near misses through our dedicated WSH Open Reporting channel. These efforts are further complemented by monthly safety posters and WSH alerts.

A WSH training roadmap had been developed for employees with WSH roles, ranging from senior management, to WSH representatives and first aiders. For example, all our WSH representatives received training for BizSAFE 2: Develop a Risk Management Implementation Plan, and championed WSH risk management within their respective divisions and departments. We organised the 3.3 and 9.9 NEA Safety Walkabouts and Safety Time-Out, during which our leaders walked the ground to conduct checks and validation, and engaged staff to gather feedback.

PLACING OUR PEOPLE FIRST

NEA creates a positive work environment for our employees through fair employment practices and work-life balance. Our management is actively involved in setting the vision and direction, formulating policies, implementing them, and ensuring a feedback loop for fair employment practices. We signed the Employers' Pledge of Fair Employment Practices in April 2007 and regularly review employment practices to align with the principles set out by the Tripartite Alliance for Fair Employment Practices. NEA's terms of employment are also guided by Singapore's legislation, including the Employment Act 1968, the Workplace Safety and Health Act 2006, and the Retirement and Re-employment Act 1993.

We are also committed to providing every individual with ample opportunities for training and career development, a balanced work life, and a safe and healthy work environment. We support the UNGC and its Ten Principles in the areas of Human Rights, Labour, Environment, and Anti-corruption.

Our employee retention rate of 93.8 per cent in 2023 has met our target to be at the 70th percentile among all statutory boards in the Public Service.

i) Employee Benefits

We value our employees and strive to provide everyone with the same set of benefits, whether they are full-time or part-time employees. Temporary staff hired via outsourced agencies would be covered by their respective agency's benefits policy.

In 2023, NEA introduced and paid a Corporate Bonus in 2023 to all our staff as another way to foster teamwork among employees and encourage them to work together to achieve key organisational goals. The payment of the Corporate Bonus is tied to performance of our corporate KPIs.

ii) Family-friendly Human Resource Policies

We support our staff in their parenthood journey by providing paid parental leave to both male and female employees. This aims to encourage shared parental responsibilities and strengthen family bonds.

Our human resource policies support employees with families and affirm our continued commitment to equality and diversity in the workplace. This commitment has resulted in a high retention rate among male and female employees who took parental leave in FY2023. Among those who returned to work after their parental leave ended, 97 per cent were still employed 12 months later¹⁷.

Return-to-work and Retention Rates of Employees Who Took Parental Leave

	Male			Female		
	FY21	FY22	FY23	FY21	FY22	FY23
Return-to-work Rate	100%	100%	100%	100%	100%	100%
Retention Rate	83%	91%	97%	84%	96%	98%

No. of Employees Who Took Parental Leave

	FY21	FY22	FY23
Female	51	56	60
Male	98	99	93

No. of Employees Who Returned to Work After Taking Parental Leave

	FY21	FY22	FY23
Female	51	56	60
Male	98	99	93

iii) Inclusive Hiring

NEA supports the hiring of persons with disabilities as part of our inclusive workplace strategy. We have implemented a holistic framework encompassing strategies, initiatives,

¹⁷ The breakdown by gender is 90 males and 59 females

and action plans to champion inclusive hiring. Additionally, we support SG Enable in its career fairs and internship programmes. As of 31 March 2024, we have created a conducive workplace for 21 persons with disabilities.

iv) Career Development

NEA offers a vibrant marketplace of internal posting opportunities to allow officers to explore new areas of NEA's diverse work and develop whole-of-NEA thinking. In addition, we provide opportunities for external stints where officers could further develop their competencies and broaden their horizons. These include short-term attachments and projects, as well as secondments to other public and private organisations.

To empower officers to take charge of their careers, NEA organises career workshops aimed at fostering a growth mindset and providing insights into the future of work. These workshops enable officers to gain a deeper understanding of their personal career preferences and explore development options available within and beyond NEA to advance their careers. NEA also offers sponsorships and post-graduate scholarships to support officers who are keen to invest in their growth and build capabilities through self-initiated learning and professional development.

Lastly, as part of NEA's efforts to grow career coaching capabilities with NEA, we have established a pool of certified career coaches led by in-service NEA officers, to support and partner staff in their career progression through various milestones.

v) Unions

Our people are our greatest assets, and we work closely with the Amalgamated Union of Public Employees and the Amalgamated Union of Public Employees - Daily-Rated Workers Branch to ensure their well-being. 93 per cent of staff are covered under the scope of collective bargaining agreements. For staff who are not covered, their terms of employment are still determined by the collective bargaining agreements set forth by the Union.

vi) Code of Conduct

NEA observes the Public Service Code of Conduct Key Principles (CoC), reflecting our zero tolerance policy on fraud and misconduct. The CoC provides guidelines concerning conflicts of interest, confidentiality, corruption prevention, and proper financial stewardship. Upholding the Public Service Values of serving with Integrity, Service and Excellence, the CoC sets out the standards of behaviour expected of all public officers. The principles guide employees in discharging their duties and responsibilities to the highest standards of personal and corporate integrity.

The CoC is available in the Staff Handbook on the NEA intranet. All new staff are informed about the CoC. In the event of a breach of discipline, the NEA Disciplinary Policy outlines the appropriate procedures to address the breach.

Employees are reminded to declare actual or potential conflicts of interest during the course of their work. When declarations are made, the relevant parties are informed. To prevent and mitigate conflicts of interest, employees are required to declare any potential conflicts between their official duties and personal interests as soon as they arise. Employees must not participate in such outside activities without CEO approval. Examples of scenarios requiring declarations include:

- Directly or indirectly using any official information or position to further a private interest
- Being involved in the activities of any organisation, whether in a personal capacity or as an office bearer of a recreational club, society, association and so on, where such activities may interfere with the objective performance of their official duties for personal advantage or gain

vii) Whistleblowing Policy

Under NEA's whistleblowing policy, employees and third parties are encouraged to report any incidents of fraud and serious misconduct. Our staff can report wrongful practices they have observed in NEA to our CEO, as stated in the Internal Disclosure Policy, which is available on the NEA Intranet.

NEA is committed to providing a safe environment for employees to report wrongful practices. All reports are treated with utmost confidence, and every effort is made to protect the identity of staff who report or provide evidence of any wrongful practice, to the extent feasible and permissible under the law.

NEA's whistleblowing channels provide stakeholders with avenues to report, in good faith and without fear of reprisal, any incidents of fraud or misconduct occurring within NEA or involving our staff, for appropriate investigation and action. These channels are independently managed by our Internal Audit Division. Any person who makes a report in good faith is assured that no action will be taken against them, even if the investigation reveals no wrongdoing. However, firm action, including reporting the matter to SPF, will be taken against anyone who makes false, malicious, or frivolous reports.

viii) Employee Grievance

NEA has implemented a grievance handling framework and is committed to working closely with the union to address and resolve staff feedback and concerns promptly. We offer employees ample channels to air their grievances on matters that affect the terms and conditions of their service. Any employee who feels aggrieved due to unfair treatment in the workplace should raise their grievance through internal official channels, either verbally or in writing.

Grievances surfaced in good faith will be treated with strict confidence and shared only for the purpose of grievance management. No officer or supervisor should retaliate against the staff who surfaced the grievance; those who do so may be subject to disciplinary action. If an allegation is made frivolously, maliciously, or for personal gain, disciplinary action may also be taken. All critical concerns are conveyed to the Chairman of the Board Audit Committee. Direct channels are available for such reporting.

EMBRACING WORKFORCE DIVERSITY

As at 31 March 2024, NEA had a diverse workforce of 3,678 working in various locations islandwide. Compared to FY2022's headcount of 3,672, the number of employees for FY2023 has remained relatively stable.

i) Breakdown by Age

NEA continued to recruit employees of all ages, with the largest proportion falling within the 30 to 50-year-old range. A significant proportion of our workforce consists of older workers. We value our employees regardless of age and actively re-employ eligible staff. Those ineligible for re-employment receive an Employee Assistance Payment and are connected with organisations such as the Employment and Employability Institute for employment and training assistance.

	FY21	FY22	FY23
No. of Employees in NEA (by Age)			
< 30 years old	459	351	334
30-50 years old	1,870	1,906	2,030
> 50 years old	1,470	1,415	1,314
Total	3,799	3,672	3,678
No. of Employees recruited (by Age)			
< 30 years old	138	111	112
30-50 years old	255	209	220
> 50 years old	82	16	17
Total	475	336	349
No. of Employees who left NEA (by Age)			
< 30 years old	87	96	50
30-50 years old	209	214	177
> 50 years old	188	153	115
Total	484	463	342

ii) Breakdown by Gender

Due to the nature of NEA's operational work, which is skewed towards enforcement and field operations, there are typically more male applicants, leading to a higher likelihood of males assuming these roles. Nevertheless, both genders receive equal consideration during the recruitment process.

	FY21	FY22	FY23
No. of Employees in NEA (by Gender)	65% 35%	65% 35%	64% 36%
Male	2,485	2,371	2,359
Female	1,314	1,301	1,319
No. of Employees recruited (by Gender)	67% 33%	56% 44%	54% 46%
Male	316	187	187
Female	159	149	162
No. of Employees who left NEA (by Gender)	67% 33%	65% 35%	60% 40%
Male	324	301	204
Female	160	162	138

iii) Breakdown by Employment Type¹⁸

The majority of our employees are employed on a full-time basis.

Breakdown of NEA Employees by Employment Type	Male	Female
Permanent and Fixed Term Contract	2,359	1,319
Temporary Contract	0	0
Non-guaranteed Hours	–	–
Full-time	2,359	1,317
Part-time	0	2

NURTURING TALENT IN NEA

Before the start of each year, all employees are encouraged to discuss their career aspirations with their supervisors during the performance management review. Employees then set career goals and development plans aligned with NEA's objectives.

NEA supports the career development plans set during the performance management review through various programmes available under the learning infrastructure for employee development.

NEA's Singapore Environment Institute collaborates with our internal and WOG stakeholders to drive learning and development initiatives for employees. These include promoting continuous learning and life-long employability, structuring on-the-job training, and leveraging technology for an employee-directed and competency-based approach to learning and development.

To promote continuous learning and lifelong employability, NEA offers our staff a series of training sessions. NEA officers are constantly honing their technical capabilities through regular workshops and courses, updated to ensure they stay relevant in our dynamic operating landscape.

To support our mature workforce, including converted monthly-rated employees, NEA partnered the Council for the Third Age for courses on topics such as digital confidence, mental wellness, and active ageing. These courses seek to equip them with knowledge and skills to manage their mental well-being while navigating a changing world.

Additionally, NEA's on-the-job training has been refined by equipping trainers and assessors with effective coaching skills. Under our Learning Fridays initiative, employees may participate in site visits, deepening appreciation of one another's contribution and strengthening the One NEA identity. This initiative also encompasses Learning Plus, covering topics from legal contracts to behavioural science and the Pair Chat application.

Through digital solutions such as LEARN¹⁹, we empower employees to take ownership of their own learning, anytime, anywhere and just-in-time. NEA also offers employees an opportunity to practise new skills, and to build their confidence within a safe and controlled environment, through the deployment of immersive learning technologies, such as the virtual reality system.

¹⁸ In FY2023, 40 non-employees (full-time equivalent) were also engaged, mainly to provide support to the various divisions. The number of non-employees engaged had decreased compared to FY2022, as NEA no longer engaged Safe Distancing Ambassadors to support the COVID-19 pandemic.

¹⁹ LEARN.gov.sg is a self-directed, digital learning platform for Singapore public officers

SUPPORTING EMPLOYEE ENGAGEMENT AND WELL-BEING

People are central to NEA's organisational performance and excellence. NEA engages our staff regularly and supports their well-being through policies and initiatives that promote work-life balance, so that they can continue to contribute in a productive, committed and purposeful manner. NEA's Employee Engagement Framework is designed to engage employees based on motivational factors.

The diverse yet interconnected nature of NEA's work requires constant effective collaboration across departments, uniting our staff as One NEA. During the #OneNEA Appreciation Week in January 2024, staff were encouraged to take this time to thank one another through e-cards and the NEA intranet's "U R Awesome" page. The NEA senior leadership team showed their appreciation to staff by serving snacks and drinks at food carnivals held at various NEA premises. Team-bonding activities were also organised.

NEA recognises the crucial role of staff mental well-being in maintaining a healthy and positive work environment. We are committed to fostering a culture that promotes mental well-being and prioritises it as part of WSH.

To enhance support and education for our staff, we rolled out a mental wellness e-newsletter, Wonders of Wellness, to provide mental wellness tips and self-care resources. NEA observed Mental Wellness Month in October to raise awareness about mental health and well-being, and organised activities such as workshops and wellness initiatives throughout the month. These efforts aimed to encourage open conversations about mental health, provide coping strategies, and highlight the significance of seeking help when needed. Wellness Ambassadors and CARE Officers²⁰ are appointed to provide staff with emotional and mental support.

To integrate mental health advocacy into staff events, we invited Beyond the Label and the Health Promotion Board to set up booths at the Food Carnival during NEA Staff Appreciation Week, and the NEA Fun Sports Carnival respectively. The latter is one of many well-received events promoting healthy and active lifestyles, alongside others such as the Urban Walk, sports clinics and tournaments. Additionally, corporate passes to popular places of interest, such as the newly opened Bird Paradise, enabled staff to recharge, unwind, and spend quality time with their loved ones.

²⁰ CARE (Caring Action in Response to Emergencies) Officers is a WOG initiative

● OUR SUSTAINABILITY COMMITMENTS

MANAGING FINANCIAL RESOURCES RESPONSIBLY

As NEA relies predominantly on government grants for funding, our daily operations must be underpinned by financially sound and sustainable decisions. To that end, NEA practises:

i) Regular Review of Fees and Charges

Fees and charges are aligned with policy objectives and periodically reviewed with these principles in mind:

- Direct cost recovery from service users
- Accounting for the full cost of goods or services provided
- Avoiding competition with the private sector
- Striving to enhance cost efficiency and affordability through process improvements

ii) Continual Review of Financial Policies

To optimise resources and uphold corporate governance standards, NEA establishes and regularly reviews financial policies and procedures. These are accessible on NEA's intranet to ensure adherence and standardisation across departments.

iii) Optimal Budget and Resource Allocation

The NEA Board approves the budget for each fiscal year, necessitating prudent spending and optimal allocation among departments based on needs. To ensure accountability and responsible budgeting, departments strive for a budget utilisation rate of 95 per cent, monitored and reported periodically to our management and MSE.

ENSURING GOOD PROCUREMENT PRACTICES

In FY2023, NEA awarded contracts to 248 suppliers, adhering to transparency, open competition, and value for money principles as outlined in the Instruction Manual on Procurement.

Aligning with national initiatives, such as the Progressive Wage Mark accreditation, NEA sources from firms that meet employment standards and pay progressive wages. A Price Quality Scoring Framework ensures objective bid evaluation, with all quotations and tenders posted on GeBIZ for transparency. Additionally, to avoid conflicts of interest, the approving authority cannot be from the procuring division.

A centralised oversight process to prioritise high-value tenders ensures prudent spending, achieves cost savings, and maintains operational efficiency. NEA has also enhanced our procurement processes through digitisation and robotic process automation, to enhance resource optimisation and financial sustainability.

ACHIEVING THE PUBLIC SECTOR'S SUSTAINABILITY TARGETS

Under the GreenGov.SG initiative which was launched in July 2021, the public sector has set sustainability targets that include reducing its energy²¹ and water²² consumption by 10 per cent by FY2030 from the average of FY2018 to FY2020 levels, reducing the amount of waste²³ disposed of by 30 per cent by FY2030 from FY2022 levels, and peaking carbon emissions around 2025 and achieving net zero emissions around 2045. NEA strives towards these targets by continuing to implement resource management measures for our premises, such as office buildings, laboratories, hawkker centres, after-death facilities, and waste-to-energy (WTE) plant.

ACHIEVING THE GREEN CERTIFICATIONS

GreenGov.SG requires all new public sector buildings to achieve at least Green Mark (GM) Platinum Super Low Energy standards or equivalent, while all existing public sector buildings are to obtain GM Platinum Super Low Energy standards or equivalent upon the next major retrofitting. NEA continually works toward GM certification for our premises, where we attained GM certifications for 11 premises. In addition, NEA Headquarters at Environment Building attained the Green Mark for Healthier Workplaces Platinum certification in May 2023.

Apart from GM certifications, we attained PUB's Water Efficient Building certification for 10 premises in FY2023. This involved the replacement or adjustment of water fittings to meet the certification's requirements. We also commenced the renewal of Eco Office certifications for our offices to ensure that environmentally friendly practices continue to be adopted.

²¹ The GreenGov.SG target for energy is to improve the energy consumption per unit of gross floor area by 10 per cent by FY2030 from the average of FY2018 to FY2020 levels

²² The GreenGov.SG target for water is to improve the water consumption per person per day by 10 per cent by FY2030 from the average of FY2018 to FY2020 levels

²³ The GreenGov.SG target for waste is to reduce the amount of waste disposed of per person per day by 30 per cent by FY2030 from FY2022 levels



ACHIEVING AWARDS WITH ENERGY-EFFICIENT DESIGN

Choa Chu Kang Columbarium, designed to facilitate resource-efficient operations while providing a beautiful and respectful space to honour the memories of the deceased, earned the second runner-up spot in the ASEAN Energy Awards 2023 for Energy Efficient Buildings – Tropical Building category.

The columbarium complex's north-south orientation promotes airflow, with naturally ventilated niche rooms that are evenly distributed to accord privacy to next-of-kin. High ceilings maximise natural light, reducing the building's reliance on electrical lighting during daylight hours.

While lighting accounts for approximately 50 per cent of electricity usage, the gradual adoption of energy-efficient LED lighting is estimated to reduce consumption by 45 per cent. The building also incorporates solar panels, which generated 196,331 kWh of electricity for internal consumption in FY2023. Additionally, excess energy is exported to the grid, and water-saving fittings are installed on washbasin taps to conserve water usage.

MANAGING CARBON EMISSIONS

Our Scope 1 emissions decreased 23.1 per cent in FY2022 compared to FY2021 due to the decommissioning of Tuas Incineration Plant in February 2022.

In FY2023, Tuas South Incineration Plant (TSIP) was the only WTE plant operated by NEA, and the main contributor to our corporate carbon footprint. Compared to FY2022, the Scope 1 emissions of the WTE plant fell by approximately seven per cent, due to a lower percentage of plastic content in the waste incinerated for the year – 24 per cent in FY2023 as compared to 27 per cent in FY2022. To further lower these Scope 1 emissions, NEA aims to reduce the amount of waste generated in Singapore and divert plastics from WTE plants through mechanical recycling initiatives. NEA has also partnered Keppel Seghers Pte Ltd to study the feasibility of implementing carbon capture technology at the WTE plants.

On the other hand, our Scope 2 emissions in FY2023 were comparable to FY2022, while Scope 3 emissions improved slightly.



Corporate Carbon Footprint²⁴

	FY21		FY22		FY23	
	Tonnes of CO ₂ e (% of total aggregate)	Ranking	Tonnes of CO ₂ e (% of total aggregate)	Ranking	Tonnes of CO ₂ e (% of total aggregate)	Ranking
Direct Carbon Emissions (Scope 1)						
WTE Plants ^{25 26}	616,401 (98.5%)	1	472,290 (98.2%)	1	438,456 (97.8%)	1
NEA Operations (e.g. landfill, vehicles) ^{27 28}	6,046 (1.0%)	2	5,727 (1.2%)	3	6,621 (1.5%)	2
Liquefied Petroleum Gas (LPG) and Town Gas ^{29 30}	2,906 (0.5%)	4	3,059 (0.6%)	4	3,000 (0.7%)	4
Sub-total CO ₂ e Emissions (Scope 1)	625,353 tonnes CO ₂ e		481,076 tonnes CO ₂ e		448,077 tonnes CO ₂ e	
Indirect Carbon Emissions (Scope 2)						
Electricity ³¹ (NEA Office Premises ³²)	2,647 (33.2%)	5	2,891 (32.9%)	5	2,872 (33.0%)	5
Electricity ³³ (NEA Facilities ³⁴)	5,330 (66.8%)	3	5,894 (67.1%)	2	5,820 (67.0%)	3
Sub-total CO ₂ e Emissions (Scope 2)	7,978 tonnes CO ₂ e		8,785 tonnes CO ₂ e		8,692 tonnes CO ₂ e	
Indirect Carbon Emissions (Scope 3) ³⁵						
Local ³⁶ Staff Business Travel	247 (78.1%)	6	239 (43.6%)	7	188 (38.9%)	7
Overseas Business Travel (Flight only)	69 (21.9%)	7	310 (56.4%)	6	295 (61.1%)	6
Sub-total CO ₂ e Emissions (Scope 3)	317 tonnes CO ₂ e		550 tonnes CO ₂ e		483 tonnes CO ₂ e	

²⁴ NEA's corporate carbon footprint is calculated in accordance with the GHG Protocol Corporate Accounting and Reporting Standard set by the World Business Council for Sustainable Development and World Resources Institute. This is in line with the 2006 Intergovernmental Panel on Climate Change (IPCC) Guidelines for National GHG Inventories, which is used in Singapore's international reporting to the UN Framework Convention on Climate Change. The computations for GHG emissions from waste incineration at TSIP are also in accordance with the 2006 IPCC Guidelines for National GHG Inventories. CO₂e, CH₄, and N₂O were included in the calculation of Scope 1 emissions

²⁵ Consumption is largely dependent on public demand for NEA's services. The Scope 1 emissions for waste incineration have been restated for FY2021 and FY2022 to reflect actual emissions. January to March 2024 figures are based on estimates, and will be finalised at the end of 2024 upon verification. The actual figure for FY2023 will be updated in next year's report

²⁶ The figures in the table are referring to the non-biogenic emissions from the WTE plants. The biogenic emissions from the plants are 715,314 tonnes CO₂, 577,003 tonnes CO₂, and 617,675 tonnes CO₂ in FY2021, FY2022, and FY2023 respectively. The biogenic emissions are not included in NEA's corporate carbon footprint to align with IPCC guideline, where biogenic emissions with CO₂ content only are reported as information item, but not included in the inventory

²⁷ Carbon emissions for NEA-owned vehicles are calculated based on amount of fuel (diesel and petrol) used. Diesel consumed by vehicles, equipment, diesel generator, and marine vessels used at Tuas Marine Transfer Station, Semakau Landfill, WTE plants were also included in the calculation of carbon emissions

²⁸ The Scope 1 emissions for NEA operations (e.g. landfill, vehicles) have been restated for FY2021 and FY2022 to 6,046 tonnes CO₂e (from 5,329 tonnes CO₂e) and 5,727 tonnes CO₂e (from 4,988 tonnes CO₂e) respectively to include more data

²⁹ Carbon emissions are calculated for LPG and Town Gas used at Choa Chu Kang Crematorium and Mandai Crematorium respectively. The emissions are impacted by factors beyond NEA's control, such as the number of after-death services required

³⁰ The Scope 1 emission figures for LPG and Town Gas for FY2021 and FY2022 have been revised to 2,906 tonnes CO₂e (from 2,316 tonnes CO₂e) and 3,059 tonnes CO₂e (from 2,358 tonnes CO₂e) respectively to use the amount of energy instead of the amount of fuel used as the input in the calculation

³¹ Grid-average emissions factor (GEF) data is used to calculate GHG emissions from electricity NEA purchased from the national grid. FY2023 figures are calculated based on 2022 electricity GEF: 0.4168 kg CO₂/kWh (<https://www.ema.gov.sg/resources/singapore-energy-statistics/chapter2>). The FY2021 and FY2022 figures have also been restated to use the updated GEF value of 0.4085 kg CO₂/kWh and 0.4168 kg CO₂/kWh

³² The Scope 2 emissions for NEA offices have been restated for FY2021 and FY2022 to 2,647 tonnes CO₂e (from 2,958 tonnes CO₂e), and 2,891 tonnes CO₂e (from 3,087 tonnes CO₂e) respectively. This was due to the revised GEF, and that our Mosquito Production Facility has been reclassified and accounted under the "NEA facilities" category

³³ The electricity consumed at TSIP and Tuas Marine Transfer Station was excluded from the carbon emissions calculation, as the electricity used at these premises is self-generated. For Semakau Landfill, the electricity consumed was generated by diesel generators on the landfill. The diesel consumed by generators had been accounted for under Scope 1 – NEA operations

³⁴ The Scope 2 emissions for NEA facilities have been restated for FY2021 and FY2022 to 5,330 tonnes CO₂ (from 1,493 tonnes CO₂) and 5,894 tonnes CO₂ (from 1,542 tonnes CO₂) respectively. This was due to the revised GEF, inclusion of Scope 2 emissions from hawkers centres, and that some premises originally accounted under "NEA offices" had been reclassified as "non-standard infrastructure" and hence accounted under the "NEA facilities" category

³⁵ The carbon emissions from local and overseas business travel have been restated to use the emission factor from Department for Environment, Food and Rural Affairs of the United Kingdom

³⁶ Carbon emissions were calculated based on local business travel by staff vehicles, excluding the travel journeys of NEA staff to and from their homes



Comparison of NEA's Corporate Emissions Intensity

	FY21	FY22	FY23
Indirect Carbon Emissions (Scope 2 & 3) (tonnes CO ₂ e) ³⁷	8,294	9,335	9,175
GFA (m ²) ³⁸	229,704.14	236,984.20	237,366.82
Carbon Emissions (tonnes CO ₂ e per person)	0.036	0.039	0.039

MONITORING AIR QUALITY

Apart from reducing our corporate carbon emissions, we also minimise the impact of our incineration and cremation activities on the environment. TSIP and Mandai Crematorium comply with Singapore's Environmental Protection and Management (Air Impurities) Regulations to manage their air emissions.

The air pollution control design features incorporated into TSIP ensure compliance with regulatory emission limits. Efficient process control systems are in place to remove pollutants in the flue gas and manage the incineration of refuse. We regularly test air emissions for both TSIP and Mandai Crematorium, to ensure that emissions are kept within limits.

Status of Air Quality Emissions

NEA's facilities	Impurities (mg/Nm ³)	FY21	FY22	FY23	Emission Limits ³⁹
Tuas South Incineration Plant	Particulates	7.5	4.3	10.8	< 50
	SO _x	92.1	125.9	85	< 1,700
	NO _x	179.1	167.4	169.3	< 400
Mandai Crematorium	Particulates	16.4	14.3	9.9	< 50
	SO _x	0.2	13.4	0.0	< 1,700
	NO _x	86.0	240.0	125.5	< 400
	HCl	9.8 ⁴⁰	36.4	2.6	< 200
	CO	5.0	5.0	5.0	< 250

CONSERVING RESOURCES WITHIN NEA

Energy Management

In FY2023, NEA's total energy consumption was 578 TJ⁴¹, with 24.6 per cent attributed to the use of liquefied petroleum gas (LPG) at Choa Chu Kang Crematorium, Town Gas at Mandai Crematorium, as well as diesel and petrol used in NEA operations, such as landfill operations and vehicle use.

NEA's facilities	Unit	FY21	FY22	FY23
Total energy consumption within NEA	TJ	658	538	578
Electricity purchased from grid, and self-generated electricity from on-site generators	TJ	522	403	431
Renewable energy, such as solar	TJ	4	4	5
LPG and Town Gas	TJ	52	55	54
Diesel and Petrol	TJ	80	76	88

Electricity Consumption

Electricity usage accounts for less than two per cent of NEA's carbon emissions. In line with GreenGov.SG, we keep track of the electricity consumption at our offices⁴² and facilities⁴³ using the two indicators:

- Total electricity consumption
- Energy Utilisation Index (EUI) – amount of electricity used per floor area

In FY2023, our total electricity consumption increased by about 7 per cent compared to FY2022. This was mainly due to an increase in electricity consumption at TSIP as the availability of boilers in operation was higher in FY2023 compared to FY2022. With the boilers in operation for a longer period, more electricity is consumed by various equipment such as the vibrating conveyors, fans and pumps for each boiler as well as refuse and ash cranes. In addition, more waste was landfilled at Semakau Landfill. Notwithstanding that, the electricity consumption at our after-death facilities and laboratories improved 8.1 per cent and 6.8 per cent respectively from FY2022's levels, while the electricity consumption of other premises types remained comparable to FY2022. This could be attributed to NEA's ongoing efforts to reduce electricity consumption through measures such as removing

³⁷ The indirect carbon emissions have been restated to include the emissions from hawker centres

³⁸ The denominator metric had been revised to use the gross floor area (GFA) instead of occupant/visitor count as our Scope 2 emissions (Scope 3 is negligible compared to Scope 2) is largely influenced by the size of the premises and less by the number of occupants/visitors

³⁹ The emission limits are stipulated by Singapore's Environmental Protection and Management (Air Impurities) Regulations

⁴⁰ The figure has been corrected to 9.8. It was previously incorrectly indicated as 98 due to typo error

⁴¹ This includes fuel consumption (LPG, Town gas, diesel, petrol and waste) + solar energy consumed onsite + energy purchased for consumption + self-generated electricity (consumed onsite)

⁴² This includes offices, laboratories, and data centres

⁴³ This includes our after-death facilities; installations such as TSIP, Semakau Landfill, and Tuas Marine Transfer Station; hawker centres, Mosquito Production Facility, and air quality and weather monitoring stations. Consumption at NEA's facilities is largely dependent on public demand for NEA's services



excess lighting, replacing lighting with LED lights, and switching our appliances to more energy-efficient models.

Electricity Consumed at NEA Offices and Facilities⁴⁴

	Unit	FY21	FY22	FY23
Total Electricity Consumed	KWh	144,937,209	111,879,534	119,757,820
Offices ⁴⁵	KWh	6,480,864	6,936,464	6,890,378
Facilities ⁴⁶	KWh	138,456,345	104,943,070	112,867,442

	Baseline ⁴⁷	FY21	FY22	FY23
EUI (KWh/m ²) ⁴⁸	83.0	76.9	81.8	80.4
Offices (KWh/m ²)	N.A.	192.6	199.2	197.1
Facilities (KWh/m ²)	N.A.	63.0	67.4	66.2

In addition, NEA deployed solar photovoltaic panels on eight⁴⁹ premises, including hawker centres, after-death facilities and Semakau Landfill. Collectively, they generated about 1,500 MWh of electricity in FY2023, of which about 1,400 MWh of solar energy was consumed on-site. We have plans to deploy solar photovoltaic panels on three after-death facilities in the coming years.

As a result, NEA's EUI in FY2023 improved by 1.8 per cent and 3.2 per cent respectively compared to FY2022 and the baseline period. NEA will continue to identify more energy reduction measures to meet the GreenGov.SG target.

Electricity Generation

The electricity generated from waste incineration is supplied to TSIP and Tuas Marine Transfer Station to support their operations. Any excess electricity from the incineration plant is exported to the power grid. The amount of electricity exported at TSIP decreased in FY2023 as compared to FY2022 as the transformers underwent servicing and repairs from May 2023 to September 2023 and the plant was disconnected from the grid. During this period, the plant remained operational and continued to receive and incinerate waste to

generate electricity for plant's internal consumption.

Electricity Generated from Waste Incineration

	Unit	FY21 ⁵⁰	FY22	FY23
Total electricity generated	MWh	545,170	400,223	277,946
Total electricity exported	MWh	421,406	310,807	180,760

Water Management

The water consumption⁵¹ at our offices and facilities is tracked using the two indicators:

- Total water consumption
- Water Efficiency Index (WEI) – amount of water used per person⁵² per day

The total water consumption in FY2023 increased by 20.4 per cent compared to FY2022. The increase in water consumption was largely due to higher amount of water consumed at TSIP due to additional maintenance works at the premises. For Choa Chu Kang Columbarium, the increase in water consumption was due to maintenance works including groundwater recharging activities. NEA also saw an increase in water consumption at our offices and hawker centres, largely due to more occupants and visitors using the facilities.

Water Consumed at NEA Offices and Facilities⁵³

	FY21	FY22	FY23
Total Water Consumed (L)	701,168,128	458,130,501	551,425,456
Offices (L)	16,778,954	17,960,701	19,372,656
Facilities (L)	684,389,174	440,169,800	532,052,800

	Baseline ⁵⁴	FY21	FY22	FY23
Overall WEI (L/person/day) ⁵⁵	29.1	26.7	23.4	26.9
Offices (L/person/day)	N.A.	11.8	16.7	14.9
Facilities (L/person/day)	N.A.	26.3	22.6	26.2

⁴⁴ This refers to the total electricity drawn from the grid and from self-generated electricity consumption

⁴⁵ To align with our GreenGov.SG reporting scope, the electricity consumption from NEA "Offices" have been restated to exclude resource consumption of the Mosquito Production Facility as it has been recategorized and reported under "Facilities" instead

⁴⁶ The electricity consumption figures for NEA facilities for FY2021 and FY2022 have been restated to include the electricity consumption from our hawker centres and the Mosquito Production Facility

⁴⁷ NEA uses the average of FY2019 to FY2020 levels as our baseline to track our EUI performance. This is in line with the overall baseline of the MSE Family, which comprises MSE, NEA, PUB and SFA

⁴⁸ The calculation of EUI excludes non-standard infrastructure such as TSIP, cremators at Mandai Crematorium, the Mosquito Production Facility, our air pollution and weather monitoring stations, Semakau Landfill, Tuas Marine Transfer Station, Kranji State Cemetery, and Inland Ash Scattering Facility

⁴⁹ NEA also deployed solar panels on four other premises, but they are not included here as the premises are excluded from GreenGov.SG scope

⁵⁰ Electricity generation and export figures for FY2021 also included contribution from Tuas Incineration Plant, which was decommissioned in 2022

⁵¹ This includes potable water and NEWater from PUB. The consumption is tracked using utility bills

⁵² This includes the number of occupants, visitors, contractors and sub-contractors who had visited NEA's premises

⁵³ Similar to electricity, the water consumption and WEI figures for FY2021 and FY2022 have been restated to include water consumption from hawker centres and reclassification of Mosquito Production Facility

⁵⁴ NEA uses the average of FY2019 to FY2020 levels as our baseline to track our WEI performance. This is in line with the MSE Family's overall baseline

⁵⁵ Similar to the EUI, the WEI is only calculated for offices and facilities, and excludes non-standard infrastructure



NEA will continue to implement water reduction measures at our premises, such as switching to 3-tick Water Efficiency Labelling Scheme (WELS) rating water fittings during repair and redecoration works. Our facility managers continued to conduct inspections at various premises, to ensure leaks are detected early and faulty hardware replaced promptly.

In FY2023, 262,015,000 L of NEWater was consumed at TSIP for purposes such as producing demineralised water for boiler usage and general washing. 92,000 L of NEWater was also consumed at Kranji State Cemetery for landscaping activities.

As part of NEA's water conservation efforts, NEA set up rain harvesting systems at several NEA premises. 48,675,170 L of rainwater was collected and used to support NEA's operations, such as general washing at TSIP and Tuas Marine Transfer Station. At the Garden of Peace inland ash scattering facility, harvested rainwater can be used by next-of-kin, with a watering can, to water the remaining ashes on the pebbles, allowing the ashes to percolate into the soil beneath.

Water Discharge

In FY2023, TSIP discharged 44,541 L of used water into public sewers. Using a flowmeter to calculate the volume of wastewater discharged into the public sewer, we monitor and measure the quality of water discharge from TSIP monthly to ensure the water discharge complies with allowable limits under PUB's Sewerage and Drainage Act (Chapter 294, Sections 72 and 74) and Sewerage and Drainage (Trade Effluent) Regulations, to minimise impact on the water bodies and related habitats.

Waste Management

Beyond initiatives to improve Singapore's waste management, NEA is committed to do our part to lengthen the lifespan of Semakau Landfill, our only landfill.

We track the waste disposed of at our offices and facilities using the two indicators:

- Total waste disposed of
- Waste Disposal Index (WDI) – amount of waste disposed of per person per day

The total waste generated⁵⁶ by NEA in FY2023 increased by 3.9 per cent compared to FY2022. The increase in waste generated was in tandem with higher footfall at NEA facilities, such as hawker centres. Overall, our WDI improved by 9.6 per cent in FY2023 compared to FY2022.

In our offices, we regularly broadcast environmentally sustainable practices. We reduce printing by reminding staff to use paper mindfully, and set all printers to print on both sides

of the paper by default. Furniture and office supplies such as files are reused to minimise wastage. Recycling bins are conveniently located at printing rooms and pantries. At our after-death facilities, we continue to incorporate messages into our notices to encourage visitors to bring back their offerings.

Overall, we collected 25,100 kg of paper, plastics, and metal recyclables at our offices and facilities, and 1,516,000 kg of metal recyclables at TSIP. A third-party recyclables contractor collects these recyclables for recovery operations off-site.

Waste Generation at NEA Offices and Facilities⁵⁷

	FY22	FY23
Total waste generated (kg)	15,070,576	15,657,426
Offices (kg)	109,616	135,358
Facilities (kg)	14,960,960	15,522,068

	Baseline (FY22)	FY23
Overall WDI (kg/person/day) ⁵⁸	1.8	1.6
Offices (kg/person/day)	0.5	0.5
Facilities (kg/person/day)	1.8	1.7

OVERSEEING BIOHAZARDOUS WASTE DISPOSAL

NEA generated biohazardous waste at a few of our premises.

In FY2023, 40,800 L of biohazardous waste and 637.6 L of chemical waste were generated from the Environmental Health Institute's laboratory work. While these could not be recycled, NEA has put in place strict disposal processes to prevent contaminating the environment. The biohazardous waste and chemical waste were stored in purpose-built waste bins before collection by licensed industrial waste collectors – Asia Medical Enviro Services Pte Ltd, Modern Asia Environmental Holdings Pte Ltd, and Chem-Solv Technologies Pte Ltd respectively. Both the biohazardous and chemical waste were then transported to a dedicated off-site incinerator for incineration.

Apart from laboratories, we also disposed of 6.9 tonnes of used oil from the TSIP, which was disposed of by our contractor, Ensure Engineering Pte Ltd, via off-site incineration.

⁵⁶ NEA collects the waste data from the public waste collectors and general waste collectors on a monthly basis. Waste generated by NEA is incinerated by a third-party waste management provider off-site, in compliance with national environmental laws and regulations on waste management

⁵⁷ The waste generation figures and WDI figures have been restated to include waste disposed of at our hawker centres

⁵⁸ Similar to the EUI and WEI, the WDI is only calculated for offices and facilities, and excludes non-standard infrastructure



GRI CONTENT INDEX



**CONTENT INDEX
ESSENTIALS SERVICE**

2024

For the Content Index – Essentials Service, GRI Services reviewed that the GRI content index has been presented in a way consistent with the requirements for reporting in accordance with the GRI Standards, and that the information in the index is clearly presented and accessible to the stakeholders.

Statement of use	NEA has reported in accordance with the GRI Standards for the period 1 April 2023 to 31 March 2024.
GRI 1 used	GRI 1: Foundation 2021
Applicable GRI Sector Standard(s)	N/A

GRI Standard / Other Source	Disclosure	Location	Page No.	Omission		Direct Answer / Additional Info
				Reasons	Explanation	
GENERAL DISCLOSURES						
GRI 2: General Disclosures 2021	2-1 Organizational details	About This Report	4	-	-	-
		About the National Environment Agency	11			
	2-2 Entities included in the organization’s sustainability reporting	About This Report	4	-	-	-
	2-3 Reporting period, frequency and contact point	About This Report	4	-	-	-
	2-4 Restatements of information	Our Sustainability Commitments	38-43	-	-	-

	2-5 External assurance	About This Report (Assurance)	4	-	-	-
	2-6 Activities, value chain and other business relationships	About This Report	4	-	-	-
		About the National Environment Agency	11			
	2-7 Employees	Living Out Earth as One NEA (Embracing Workforce Diversity)	35-36	-	-	-
	2-8 Workers who are not employees	Living Out Earth as One NEA (Embracing Workforce Diversity – Breakdown by Employment Type)	35	-	-	-
	2-9 Governance structure and composition	Governance	9-10	-	-	-
	2-10 Nomination and selection of the highest governance body	Governance	9-10	-	-	-
	2-11 Chair of the highest governance body	Governance	9-10	-	-	-
	2-12 Role of the highest governance body in overseeing the management of impacts	Governance	9-10	-	-	-

	2-13 Delegation of responsibility for managing impacts	Governance	9-10	-	-	-
	2-14 Role of the highest governance body in sustainability reporting	Governance About the National Environment Agency (Enterprise Risk Management)	9-10 15	-	-	GRI 2-14a: The Board oversees NEA's sustainability plans, strategies and targets, which feed into the sustainability reporting process that is driven by our senior management. The Annual & Sustainability Report is subsequently reviewed and approved by the Minister for Sustainability and the Environment before its launch.
	2-15 Conflicts of interest	Governance	9-10	-	-	-
	2-16 Communication of critical concerns	Living Out Earth as One NEA (Placing Our People First – Whistleblowing Policy, Employee Grievance)	33-35	GRI 2-16a: Confidentiality constraints	One critical concern was raised to the approving authority. However, the nature of the concern is not disclosed due to sensitivity concerns.	-
	2-17 Collective knowledge of the highest governance body	-	-	Not applicable	NEA's mission centres on sustainability, and the Board provides strategic direction towards our mission. Hence, the Board is familiar with sustainability topics.	-

	2-18 Evaluation of the performance of the highest governance body	-	-	Confidentiality constraints	This will not be disclosed as the evaluation process contains sensitive information of business affairs, since NEA is a statutory board.	-
	2-19 Remuneration policies	-	-	Confidentiality constraints	NEA's compensation policies are aligned to the guidelines of Public Service Division. As the policies are confidential, no further details will be shared.	-
	2-20 Process to determine remuneration	-	-	Confidentiality constraints	NEA's compensation policies are aligned to the guidelines of Public Service Division. As the policies are confidential, no further details will be shared.	-
	2-21 Annual total compensation ratio	-	-	Confidentiality constraints	NEA's compensation policies are aligned to the guidelines of Public Service Division. As the policies are confidential, no further details will be shared.	-
	2-22 Statement on sustainable development strategy	Chairman's Foreword CEO's Message	5 6	-	-	-
	2-23 Policy commitments	Living Out Earth as One NEA Our Sustainability Commitments	30-37 38-43	-	-	-

	2-24 Embedding policy commitments	Living Out Earth as One NEA Our Sustainability Commitments	30-37 38-43	-	-	-
	2-25 Processes to remediate negative impacts	Living Out Earth as One NEA (Placing Our People First – Whistleblowing Policy, Employee Grievance)	34-35	-	-	-
	2-26 Mechanisms for seeking advice and raising concerns	Living Out Earth as One NEA (Placing Our People First – Whistleblowing Policy)	34	-	-	-
	2-27 Compliance with laws and regulations	-	-	Confidentiality constraints	As information on the Tuas Incineration Plant incident is presently before the Courts, we are not able to provide further information.	-
	2-28 Membership associations	About the National Environment Agency	11	-	-	-
	2-29 Approach to stakeholder engagement	About the National Environment Agency (Material Issues) Upholding High Public Health Standards (Adjusting the Orchard Road No-Smoking Zone, Reining in Rat Burrow Numbers, Shaving the Dengue Peak)	12-14 18-19	-	-	-

		Safeguarding Singapore's Environment	21			
		Nurturing a Sustainable Singapore (Implementing Stringent Standards for Carbon Offsets)	27			
		Living Out Earth as One NEA (Fostering international Partnerships and Collaborations, Engaging our Partners and Community, Embarking on Operational Reviews, Nurturing Talent in NEA)	31-32, 36			
	2-30 Collective bargaining agreements	Living Out Earth as One NEA (Placing Our People First)	34	-	-	-
MATERIAL TOPICS						
GRI 3: Material Topics 2021	3-1 Process to determine material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
	3-2 List of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-

HIGH PUBLIC HEALTH STANDARDS						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
		Upholding High Public Health Standards	17-20			
SAFE, HEALTHY AND CONDUCIVE LIVING ENVIRONMENT						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
		Safeguarding Singapore’s Environment (Keeping Tabs on Air Quality, Monitoring Recreational Water Quality)	21-22			
		Our Sustainability Commitments (Monitoring Air Quality)	41			
		Our Sustainability Commitments (Conserving Resources within NEA – Water Management)	42-43			
GRI 303: Water and Effluents 2018	303-1 Interactions with water as a shared resource	Upholding High Public Health Standards (Automating Waterway Maintenance)	17	-	-	No water bodies and related habitats were significantly affected by water withdrawal, discharges or runoff.

		Safeguarding Singapore's Environment (Monitoring Recreational Water Quality)	22			
		Our Sustainability Commitments (Achieving the Green Certifications)	38			
		Our Sustainability Commitments (Conserving Resources within NEA – Water Management)	42-43			
	303-2 Management of water discharge-related impacts	Our Sustainability Commitments (Conserving Resources within NEA – Water Discharge)	43	-	-	No water bodies and related habitats were significantly affected by water discharges.
	303-3 Water withdrawal	Our Sustainability Commitments (Conserving Resources within NEA – Water Management)	42-43	-	-	GRI 303-3d: Water withdrawal is from third-party sources – PUB, and tracked using water meters.
	303-4 Water discharge	Our Sustainability Commitments (Conserving Resources within NEA – Water Discharge)	43	GRI 303-4a: Information unavailable / incomplete	NEA has disclosed water discharge information for TSIP. Water discharge information for other facilities and sites are unavailable. We aim to report on our progress in the future.	-

	303-5 Water consumption	Our Sustainability Commitments (Conserving Resources within NEA – Water Management)	42-43	-	-	GRI 303-5b: NEA does not consume water from water stress areas.
GRI 305: Emissions 2016	305-7 Nitrogen oxides (NO _x), sulfur oxides (SO _x), and other significant air emissions	Safeguarding Singapore’s Environment (Keeping Tabs on Air Quality, Monitoring Recreational Water Quality)	21-22	-	-	GRI 305-7a: NEA only tracks NO _x , SO _x , particulate matter (dust), and other significant air emissions such as CO and HCl, as these are relevant to NEA’s operations.
		Our Sustainability Commitments (Monitoring Air Quality)	41			
SUSTAINABLE AND RESOURCE EFFICIENT SINGAPORE						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
		Nurturing a Sustainable Singapore	25-27			
GRI 302: Energy 2016	302-1 Energy consumption within the organization	Our Sustainability Commitments (Conserving Resources within NEA – Energy Management)	41	-	-	-
	302-3 Energy intensity	Our Sustainability Commitments (Conserving Resources within NEA – Energy Management)	41-42	-	-	-

	302-4 Reduction of energy consumption	Our Sustainability Commitments (Conserving Resources within NEA – Energy Management)	41	-	-	-
	302-5 Reductions in energy requirements of products and services	Our Sustainability Commitments (Conserving Resources within NEA – Energy Management)	41	-	-	-
GRI 305: Emissions 2016	305-1 Direct (Scope 1) GHG emissions	Our Sustainability Commitments (Managing Carbon Emissions)	39-40	-	-	-
	305-2 Energy indirect (Scope 2) GHG emissions	Our Sustainability Commitments (Managing Carbon Emissions)	39-41	-	-	-
	305-3 Other indirect (Scope 3) GHG emissions	Our Sustainability Commitments (Managing Carbon Emissions)	39-41	-	-	-
	305-4 GHG emissions intensity	Our Sustainability Commitments (Managing Carbon Emissions)	39-41	-	-	-
	305-5 Reduction of GHG emissions	Our Sustainability Commitments (Managing Carbon Emissions)	39-41	-	-	-

GRI 306: Waste 2020	306-1 Waste generation and significant waste-related impacts	Our Sustainability Commitments (Conserving Resources within NEA – Waste Management)	43	-	-	-
	306-2 Management of significant waste-related impacts	Our Sustainability Commitments (Conserving Resources within NEA – Waste Management)	43	-	-	NEA does not import or export hazardous waste.
	306-3 Waste generated	Our Sustainability Commitments (Conserving Resources within NEA – Waste Management)	43	-	-	-
	306-4 Waste diverted from disposal	Our Sustainability Commitments (Conserving Resources within NEA – Waste Management)	43	-	-	-
	306-5 Waste directed to disposal	Our Sustainability Commitments (Conserving Resources within NEA – Waste Management)	43	-	-	-
TIMELY AND RELIABLE WEATHER AND CLIMATE SERVICES						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-

	3-3 Management of material topics	Delivering Timely and Reliable Weather and Climate Services	28	-	-	-
VIBRANT HAWKER CULTURE						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
		Cherishing our Vibrant Hawker Culture	29			
PRIVATE SECTOR AS FORCE MULTIPLIERS FOR MISSION SUCCESS						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
		Living Out Earth as One NEA (Engaging our Partners and Community)	31			
ACTIVE COMMUNITY AS STEWARDS OF THE ENVIRONMENT						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
		Nurturing a Sustainable Singapore (Empowering Zero Waste Champions in Workplaces, Striving for Zero Waste, Encouraging Recycling in Households, Promoting Green Living with Climate Vouchers)	25-26			

PUBLIC SECTOR TAKING THE LEAD IN ENVIRONMENTAL SUSTAINABILITY						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
		Safeguarding Singapore’s Environment	21-24			
INTERNATIONAL PARTNERSHIPS AS STRATEGIC LEVERS						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
		Upholding High Public Health Standards (Shaving the Dengue Peak – Collaborating with Stakeholders, Building International Capabilities for Disease Control)	18-19			
		Safeguarding Singapore’s Environment (Enhancing Regional Nuclear Safety, Security, and Safeguards, Showcasing Radiation Expertise on the International Stage, Maintaining Excellence in International Radiochemistry)	23-24			

		Living Out Earth as One NEA (Fostering International Partnerships and Collaboration)	31	-	-	-
CREATION OF CAPACITY						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
		Upholding Public Health Safety (Automating Waterway Maintenance, Raising the Bar for Cleaning Businesses, Reining in Rat Burrow Numbers, Flushing Out Zika with Wastewater Surveillance)	17-19			
		Safeguarding Singapore’s Environment (Advancing Radiation Control with Technology)	24			
		Living Out Earth as One NEA (Pursuing Excellence and Innovation to Create Capacity)	30			
STRONG CAPABILITIES FOR MISSION SUCCESS						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-

		Upholding High Public Health Standards (Building International Capabilities for Disease Control, Detecting and Intercepting Hazards)	19-20	-	-	-
		Safeguarding Singapore's Environment (Advancing Radiation Control with Technology)	24			
		Nurturing a Sustainable Singapore (Investing in Waste and Resource Research and Innovation)	25-26			
		Delivering Timely and Reliable Weather and Climate Services	28			
		Living Out Earth as One NEA (Pursuing Excellence and Innovation to Create Capacity)	30-31			
STRONG CORPORATE PRACTICES FOR A SUSTAINABLE ORGANISATION						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
GRI 201: Economic Performance 2016	201-1 Direct economic value generated and distributed	Refer to Annual Financial Statements	Pg 5 of Financial Statements	-	-	-

	201-2 Financial implications and other risks and opportunities due to climate change	Delivering Timely and Reliable Weather and Climate Services	28	-	-	-
	201-3 Defined benefit plan obligations and other retirement plans	Refer to Annual Financial Statements	Pg 13 of Financial Statements	-	-	-
	201-4 Financial assistance received from government	Our Sustainability Commitments (Managing Financial Resources Sustainably)	38	-	-	See Annual Financial Statements pages 5 and 9 for further details.
GRI 401: Employment 2016	401-1 New employee hires and employee turnover	Living Out Earth as One NEA (Embracing Workforce Diversity)	35-36	-	-	-
	401-2 Benefits provided to full-time employees that are not provided to temporary or part-time employees	Living Out Earth as One NEA (Placing Our People First)	33-34	-	-	-
	401-3 Parental leave	Living Out Earth as One NEA (Placing Our People First)	33-34	-	-	-
GRI 403: Occupational Health and Safety 2018	403-1 Occupational health and safety management system	Living Out Earth as One NEA (Prioritising Workplace Safety and Health)	32-33	-	-	-

	403-2 Hazard identification, risk assessment, and incident investigation	Living Out Earth as One NEA (Prioritising Workplace Safety and Health)	32-33	-	-	-
	403-3 Occupational health services	Living Out Earth as One NEA (Prioritising Workplace Safety and Health)	32-33	-	-	-
	403-4 Worker participation, consultation, and communication on occupational health and safety	Living Out Earth as One NEA (Prioritising Workplace Safety and Health)	32-33	-	-	-
	403-5 Worker training on occupational health and safety	Living Out Earth as One NEA (Prioritising Workplace Safety and Health)	32-33	-	-	-
	403-6 Promotion of worker health	Living Out Earth as One NEA (Prioritising Workplace Safety and Health)	32-33	-	-	-
	403-7 Prevention and mitigation of occupational health and safety impacts directly linked by business relationships	Living Out Earth as One NEA (Prioritising Workplace Safety and Health)	32-33	-	-	-

	403-8 Workers covered by an occupational health and safety management system	Living Out Earth as One NEA (Prioritising Workplace Safety and Health)	32-33	-	-	-
	403-9 Work-related injuries	Living Out Earth as One NEA (Prioritising Workplace Safety and Health)	32-33	-	-	GRI 403-9b: NEA aligns with the Ministry of Manpower in using the injury rate per 100,000 workers to assess WSH performance. NEA has not tracked the number of hours worked and work-related injuries among contractors. GRI 403-9c: No work-related hazards with risks of high-consequence injury were identified in FY23.
CONDUCTIVE CULTURE FOR COLLABORATION, LEARNING AND INNOVATION						
GRI 3: Material Topics 2021	3-3 Management of material topics	About the National Environment Agency (Material Issues)	12-14	-	-	-
GRI 404: Training and Education 2016	404-1 Average hours of training per year per employee	Living Out Earth as One NEA (Nurturing Talent in NEA)		Not applicable	NEA has pivoted towards competency-based learning to focus on both job-specific and organisational competencies, rather than the number of learning hours.	-
	404-2 Programs for upgrading employee skills and transition assistance programs	Living Out Earth as One NEA (Embracing Workforce Diversity, Nurturing Talent in NEA)	35-36	-	-	-



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